PHYSICS

1) Find the torque $(\vec{\tau} = \vec{r} \times \vec{F})$ of a force $\vec{F} = -3\hat{i} + \hat{j} + 5\hat{k}$ acting at the point $\vec{r} = 7\hat{i} + 3\hat{j} + \hat{k}$

(1) $14\hat{i} - 38\hat{j} + 16\hat{k}$

(2) $4\hat{i} + 4\hat{j} + 6\hat{k}$

(3) $-14\hat{i} + 38\hat{j} - 16\hat{k}$

(4) $-21\hat{i} + 3\hat{j} - 5\hat{k}$

2) Find the projection of $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ on $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$

(1) $\frac{5}{\sqrt{6}}$

(2) $\frac{7}{10}$

(3) $\frac{6}{\sqrt{5}}$

(4) $\frac{5}{\sqrt{3}}$

3) If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - 4\vec{b}$ are perpendicular to each other then the angle between $\vec{a}$ and $\vec{b}$ is

(1) 45°

(2) 60°

(3) $\cos^{-1}\left(\frac{1}{3}\right)$

(4) $\cos^{-1}\left(\frac{2}{7}\right)$

4) $\vec{A}, \vec{B}$ and $\vec{C}$ are vectors each having a unit magnitude. If $\vec{A} + \vec{B} + \vec{C} = \vec{0}$, then $\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C}$ will be :-

(1) – 1

(2) $-\frac{3}{2}$

(3) $-\frac{1}{2}$

(4) 0

5) The scalar product of two vectors is $2\sqrt{3}$ and the magnitude of their vector product is 2. The angle between them is :

(1) 30º
(2) 60º
(3) 45º
(4) 90º

6) An expression which cannot be defined meaningfully among vectors is :

(1) $\vec{A}.(\vec{B} \times \vec{C})$
(2) $\vec{A} \times (\vec{B} \times \vec{C})$
(3) $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D})$
(4) $(\vec{A}.\vec{B}) \times (\vec{C}.\vec{D})$

7) If $\vec{a} = 2\hat{i} + 6\hat{j} + m\hat{k}$ and $\vec{b} = n\hat{i} + 12\hat{j} + 4\hat{k}$ are parallel to each other, then product of m and n is :-

(1) 2
(2) 4
(3) 8
(4) 16

8) Vectors which is/are perpendicular to $a\cos\theta\hat{i} + b\sin\theta\hat{j}$ is :-

(1) $b\sin\theta\hat{i} - a\cos\theta\hat{j}$
(2) $\frac{\sin\theta\hat{i}}{a} - \frac{\cos\theta\hat{j}}{b}$
(3) $5a\hat{k}$
(4) All of above

9) If $|\vec{A} \times \vec{B}| = 4$ and $|\vec{A} \cdot \vec{B}| = 3$ find the value of $(\vec{A} + \vec{B})$:

(1) $\sqrt{A^2 + B^2 + \frac{8AB}{5}}$
(2) $\sqrt{A^2 + B^2 + \frac{6AB}{5}}$
(3) $\sqrt{A^2 + B^2 - \frac{8AB}{5}}$
(4) $\sqrt{A^2 + B^2 - \frac{6AB}{5}}$

10) Under a force $\left(10\hat{i} - 3\hat{j} + 6\hat{k}\right)$ newton, a body of mass 5kg moves from position $\left(6\hat{i} + 5\hat{j} - 3\hat{k}\right)$m to position $\left(10\hat{i} - 2\hat{j} + 7\hat{k}\right)$m. Find the work done

(1) 79 Joule

(2) 100 Joule

(3) 89 Joule

(4) 121 Joule

11) The cross product of the vectors $(2\hat{i}-3\hat{j}+4\hat{k})$ and $(\hat{i}+4\hat{j}-5\hat{k})$ is

(1) $-\hat{i}-14\hat{j}+11\hat{k}$

(2) $-\hat{i}-14\hat{j}+5\hat{k}$

(3) $-\hat{i}+14\hat{j}+11\hat{k}$

(4) $-\hat{i}+14\hat{j}+5\hat{k}$

12) The area of a parallelogram whose diagonals are $\vec{P}=2\hat{i}+3\hat{j}$ and $\vec{Q}=\hat{i}+4\hat{j}$ is :-

(1) 5 unit

(2) 10 unit

(3) 20 unit

(4) 2.5 unit

13) A proton in a cyclotron changes its velocity from 30 km/sec north to 40 km/sec east in 20sec. What is the magnitude of average acceleration during this time ?

(1) 0.5 km/sec

(2) 3.5 km/sec

(3) 2.5 km/sec

(4) 12.5 km/sec

14) An object travels 10 km at a speed of 100 m/s and another 10 km at 50 m/s. The average speed over the whole distance is :-

(1) 75 m/s

(2) 55 m/s

(3) 66.7 m/s

(4) 33.3 m/s

15) A car travels a distance A to B at a speed of 40 km/h and returns to A at a speed of 30 km/h. (i) What is the average speed for the whole journey? (ii) What is the average velocity?

(1) 34.3 km/hr., 0

(2) 44.3 km/hr., 0

(3) 34.3 km/hr., 34.3 km/hr

(4) none of these

16)

A particle moves along a straight line such that its displacement at any time t is given by :-

$s = t^3 - 6t^2 + 3t + 4$ meters

The velocity when the acceleration is zero is :-

(1) 3 ms^{-1}

(2) –12 ms^{-1}

(3) 42 ms^{-1}

(4) –9 ms^{-1}

17) For motion of an object along the x-axis the velocity v depends on the displacement x as $v = 3x^2 - 2x$, then what is the acceleration at x = 2m.

(1) 48 ms^{-2}

(2) 80 ms^{-2}

(3) 18 ms^{-2}

(4) 10 ms^{-2}

18) A body travels for 15 sec. starting from rest with constant acceleration. If it travels distance S_1, S_2 and S_3 in the first 5 sec., second 5 sec. and next 5 sec. respectively the relation between S_1, S_2 and S_3 is :-

(1) $S_1 = S_2 = S_3$

(2) $5S_1 = 3S_2 = S_3$

(3) $S_1 = \frac{S_2}{3} = \frac{S_3}{5}$

(4) $S_1 = \frac{S_2}{5} = \frac{S_3}{3}$

19) A particle has an initial velocity of $(3\hat{i} + 4\hat{j})$ m/s and a constant acceleration of $(4\hat{i} - 3\hat{j})$ m/s^2. Its speed after one second will be equal to :-

(1) 0

(2) 10 m/s

(3) $5\sqrt{2}$ m/s

(4) 25 m/s

20) A balloonist releases a bag from a balloon rising constantly at 40 ms^{-1} at a time when the balloon is 100 m above the ground. If g = 10 ms^{-2}, then the bag reaches the ground in :-

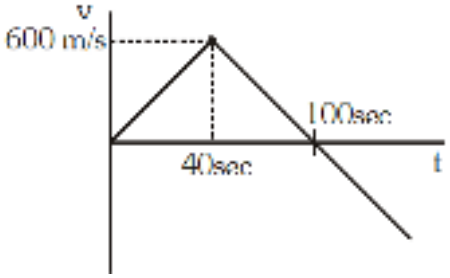
(1) 16 s^{-1}

(2) 18 s

(3) 10 s

(4) 20 s

21) A rocket is projected upwards. Whose velocity-time graph is shown. The maximum height attained by rocket from ground is

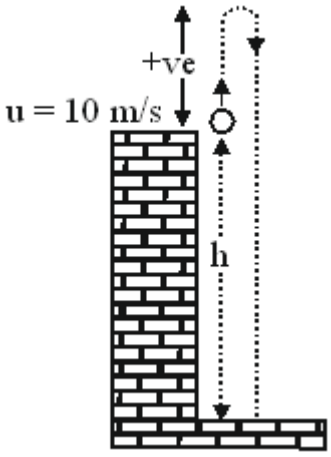


(1) 80 km
(2) 12 km
(3) 20 km
(4) 30 km

22) A body falling from height 'h' takes t_1 time to reach the ground. The time t_2 taken to cover the first half of the height is :

(1) $t_2 = \frac{t_1}{\sqrt{2}}$
(2) $t_1 = \frac{t_2}{\sqrt{2}}$
(3) $t_2 = \sqrt{3}\, t_1$
(4) None of these

23) A ball is thrown upwards from the top of a tower 40 m high with a velocity of 10 m/s, the time when it strikes the ground ($g = 10\ m/s^2$)



(1) 1s
(2) 2s
(3) 3s
(4) 4s

24)

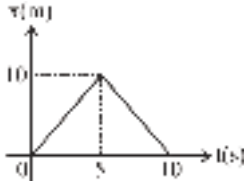
A body dropped from the top of the tower covers a distance 9x in the last second of its journey, where x is the distance covered in first second. How much time does it take to reach the ground ?

(1) 3s

(2) 4s

(3) 5s

(4) 6s

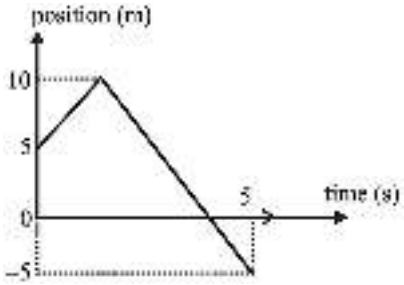
25) For a given position time graph
(for1-dimensional motion) the average speed and average velocity (for whole duration) are
respectively : 

(1) 0 ms^{-1}, 0 ms^{-1}

(2) 1 ms^{-1}, 1 ms^{-1}

(3) 2 ms^{-1}, 2 ms^{-1}

(4) 2 ms^{-1}, 0 ms^{-1}



26) Average velocity of the particle in time t = 0 to t = 5 s is :-

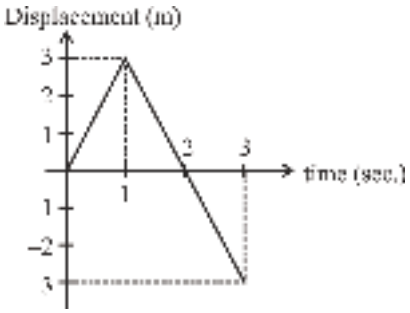
(1) 2 m/s

(2) – 2 m/s

(3) 1 m/s

(4) – 1 m/s

27) The diagram shows the displacement-time graph for a particle moving in a straight line. Find the average speed for the interval from t = 0 to t = 3 sec.



(1) 1.5 m/s

(2) 3 m/s

(3) 1 m/s

(4) 4.5 m/s

28) The velocity of the particle at any time t is given by $v = 2t(3 - t)\ ms^{-1}$. At what time is its velocity is maximum?

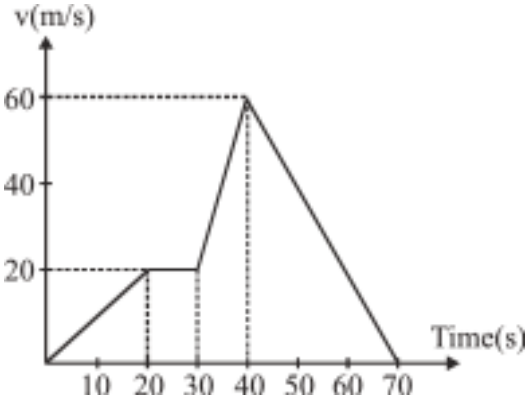
(1) 2s

(2) 3s

(3) $\frac{2}{3}s$

(4) $\frac{3}{2}s$

29) The velocity-time graph of a body is given below find average velocity from t = 0 to t = 40 sec. :-



(1) 20 m/s

(2) 40 m/s

(3) 50 m/s

(4) 60 m/s

30)

If velocity of particle is $v = \sqrt{x+2}$ m/sec, find out acceleration of particle at x = 2 m.

(1) 1 m/sec^2

(2) 0.5 m/sec^2

(3) 2 m/sec^2

(4) 4 m/sec^2

31) A ball is thrown downwards with velocity v from the top of a tower and it reaches the ground with speed 3v. What is the height of the tower :-

(1) $\frac{v^2}{g}$

(2) $\frac{2v^2}{g}$

(3) $\frac{4v^2}{g}$

(4) $\frac{8v^2}{g}$

32) A stone is dropped from the top of the building of height 80m. Calculate velocity of stone just before striking the ground :-

(1) 80 m/s
(2) 40 m/s
(3) 160 m/s
(4) 20 m/s

33) A rocket is fired vertically from the ground. It moves upwards with a constant acceleration 20 m/s^2 for 30 sec, after which the fuel is finished. After what time from the instant of firing the rocket will attain the maximum height ? (Take g = 10 m/s^2)

(1) 90 s
(2) 45 s
(3) 60 s
(4) 75 s

34) A car moving with speed 15 m/s. Suddenly the driver sees an obstruction on the road and takes 0.4 sec to apply the brake, which cause a deacceleration of 5 m/s^2. Then distance travelled by car before stopping.

(1) 22.5 m
(2) 24.5 m
(3) 26.5 m
(4) 28.5 m

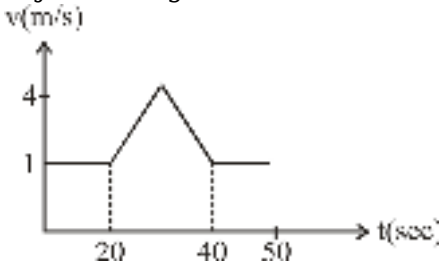
35) A truck travelling towards North with velocity 20 m/s turns East and travels at the same speed. The change in its velocity is :-

(1) $20\sqrt{2}$ m/s North-East
(2) $20\sqrt{2}$ m/s South-East
(3) $40\sqrt{2}$ m/s North-East
(4) $\frac{20}{\sqrt{2}}$ m/s North-West

36) The X and Y coordinates of the particle at any time are $x = 5t - 2t^2$ and $y = 10t$ respectively where X and Y are in metres and t is in second then the acceleration of the particle at t = 2 second is :

(1) 5 m/s^2
(2) –4 m/s^2
(3) –8 m/s^2
(4) 0 m/s^2

37) Velocity time graph for a moving object is shown in the figure then total displacement of the object during the time interval when there is non zero acceleration and retardation is :



(1) 60 m

(2) 50 m

(3) 30 m

(4) 40 m

38) If acceleration of particle is a = (2x + 1) m/sec^2 find out velocity of particle at x = 3m. If velocity at origin is 4 m/sec. :-

(1) $\sqrt{28}$ m/sec

(2) $\sqrt{90}$ m/sec

(3) $\sqrt{40}$ m/sec

(4) $\sqrt{60}$ m/sec

39) A car covers first half distance with speed v_0 and remaining half with speed $5v_0$, average speed of car is :-

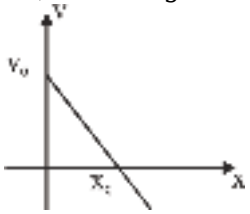
(1) $\frac{3v_0}{5}$

(2) $\frac{5v_0}{3}$

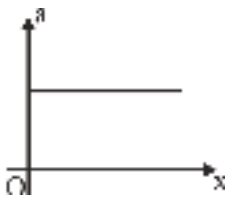
(3) $3v_0$

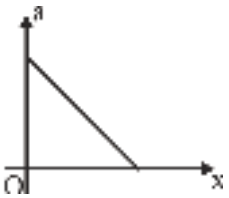
(4) $\frac{3v_0}{2}$

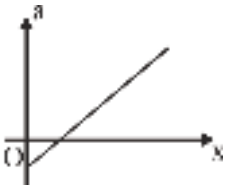
40) For the given velocity-displacement (v-x) graph, correct acceleration displacement (a-x) graph is

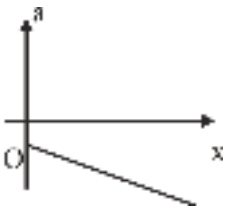


(1)



(2) 

(3) 

(4) 

41) A particle is thrown vertically upwards. Its velocity at half of maximum height is 10 m/s. The maximum height attained by it is :-

(1) 8 m
(2) 20 m
(3) 10 m
(4) 16 m

42) A particle is dropped from a tower. It is found that it travles 45 m in the last second of its journery. Find out the height of the tower :-

(1) 80 m
(2) 120 m
(3) 125 m
(4) 180 m

43) A stone falls freely such that the distance covered by it in the last second of its motion is equal to the distance covered by it in the first 5 sec. It remained in air for :-

(1) 12 sec
(2) 26 sec
(3) 13 sec
(4) 25 sec

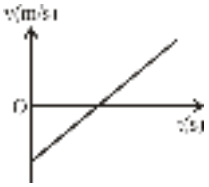
44) A body is released from the top of a tower of height H metres . It takes t time to reach the ground. Where is the body $\frac{t}{4}$ time after the release :

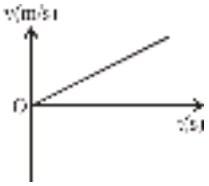
(1) At $\frac{H}{2}$ metres from ground
(2) At $\frac{15H}{16}$ metres from ground

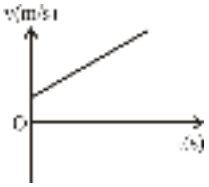
(3) At $\frac{3H}{4}$ metres from the ground

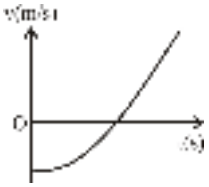
(4) At $\frac{H}{6}$ metres from the ground

45) A particle moves along x-axis in such a way that its x-coordinate varies with time t according to the equation
$x = (8 - 4t + 6t^2)$ metre.The velocity of the particle will vary with time according to the graph :-

(1) 

(2) 

(3) 

(4) 

CHEMISTRY

1) Find the **incorrect** relation?

(1) 1 amu = 1.67×10^{-24} g

(2) 1 eV = 1.6×10^{-19} J

(3) 1 erg = 10^{7} J

(4) 1 atm = 760 torr

2) The number of moles of water present in 90 g of water is :-

(1) 2

(2) 3

(3) 4

(4) 5

3) Which of the following has maximum no. of molecules ?

(1) 48 g O_3

(2) 44 g CO_2

(3) 64 g SO_2

(4) 8 g H_2

4) One litre of a gas at STP weighs 1.25 g. It can possibly be :-

(1) C_2H_2

(2) CO

(3) O_2

(4) CH_4

5) The number of atoms present in one mole H_2SO_4 would be :-

(1) $3 \times 6.02 \times 10^{23}$

(2) 6.022×10^{23}

(3) $7 \times 6.02 \times 10^{23}$

(4) $6 \times 6.022 \times 10^{23}$

6) Actual weight of a molecule of $CaCO_3$ is :

(1) 1.67×10^{-24} g

(2) 1.67×10^{-26} g

(3) 1.67×10^{-22} g

(4) 1.67×10^{-20} g

7) A gas is found to have the formula $(S_2)_X$. It's vapour density is 128. Find the value of X :-
(Molecular weight of S_2 = 64 amu)

(1) 1

(2) 2

(3) 3

(4) 4

8) The empirical formula of $C_8H_{12}O_4$ is :-

(1) $C_8H_{12}O_4$

(2) $C_4H_6O_2$

(3) C_2H_3O

(4) All

9) A compound contains 11.5% by mass of sodium. Its minimum molecular weight will be :-

(1) 80 amu

(2) 200 amu

(3) 250 amu

(4) 300 amu

10) An oxide of sulphur contains 40% by mass of sulphur. Its empirical formula is :-

(1) SO_2

(2) SO_3

(3) S_2O_3

(4) S_2O

11) The number of atoms of Cr and O are 4.8×10^{10} and 9.6×10^{10} respectively. Its empirical formula is:

(1) Cr_2O_3

(2) CrO_2

(3) Cr_2O_4

(4) None

12) If vapour density of a gas is 50. Then calculate molecular weight of gas :

(1) 25

(2) 100

(3) 200

(4) 150

13)

1.5 moles of O_2 combine with Mg to form MgO. The moles of Mg that has combined is

$2Mg + O_2 \rightarrow 2MgO$

(1) 3 mol

(2) 1.5 mol

(3) 2 mol

(4) 1 mol

14)

Among the following choose the ones with equal number of atoms.
A. 212 g of Na_2CO_3 (s) [molar mass = 106 g]
B. 248 g of Na_2O (s) [molar mass = 62 g]
C. 240 g of NaOH (s) [molar mass = 40 g]
D. 12 g of H_2 (g) [molar mass = 2 g]
E. 220 g of CO_2 (g) [molar mass = 44 g]
Choose the **correct** answer from the options given below :

(1) A, B and C only

(2) A, B and D only

(3) B, C and D only

(4) B, D and E only

15) **Statement-I :** A molecule of butane, C_4H_{10} has a mass of 58 amu.
Statement-II : One mole of butane contains 6.022×10^{23} molecules and has a mass of 58g.

(1) Statement-I is correct while statement-II is incorrect

(2) Both statement-I and statement-II are correct

(3) Both Statement-I and statement-II are incorrect

(4) Statement-I is incorrect and statement-II is correct

16) The volume of CO_2 at NTP obtained by heating 10g $CaCO_3$ will be :-

(1) 224 L

(2) 2.24 L

(3) 22.4 L

(4) 0.224 L

17) If 6×10^{22} molecules are removed from 120 g of $CaCO_3$. then find out number of moles of $CaCO_3$ left.

(1) 1.3 mol

(2) 1.2 mol

(3) 1 mol

(4) 1.1 mol

18)

What mass of oxygen is required in the complete combustion of 450 g of ethane (C_2H_6)?

(1) 3 kg

(2) 1.68 kg

(3) 1.12 kg

(4) 3.36 kg

19) Which of the following has the highest mass ?

(1) 1 g atom of C

(2) $\frac{1}{2}$ mole of CH_4

(3) 10 mL of water

(4) 3.011×10^{23} atoms of oxygen

20) The reaction $2C + O_2 \rightarrow 2CO$ is carried out by taking 24 g of carbon and 96 g of O_2. Which one is the limiting reagent?

(1) C

(2) O_2

(3) CO_2

(4) None

21) The number of electrons in 14 g N^{-3}

(1) 10

(2) 6.023×10^{23}

(3) 6.023×10^{24}

(4) 14

22) Match the column :-

Column-I		**Column-II**	
(P)	8 g of O_2	(A)	2 mole
(Q)	4 g of H_2	(B)	13 mole
(R)	52 g of He	(C)	0.25 mole
(S)	35.5 g of Cl_2	(D)	0.5 mole

Correct match will be :-

(1) P-C, Q-A, R-B, S-D

(2) P-C, Q-A, R-D, S-B

(3) P-D, Q-A, R-B, S-C

(4) P-B, Q-A, R-D, S-C

23) What is the molarity of solution in which 5.85 g of NaCl present in 20 mL of solution :-

(1) 5 M

(2) 2 M

(3) $\frac{1}{200}$M

(4) $\frac{1}{50}$M

24) Find the volume of CO_2 obtained at S.T.P. on heating 200 g of 50% pure $CaCO_3$ -

(1) 11.2 litre

(2) 22.4 litre

(3) 44.8 litre

(4) 5.6 litre

25) Which of the following is not an example of endothermic reaction :

(1) Dissociation

(2) Fusion

(3) Sublimation
(4) Combustion

26) Which of the following is not a characteristics of equilibrium :

(1) Rate is equal in both direction
(2) Measurable quantities are constant at equilibrium
(3) Equilibrium occurs in reversible condition
(4) Equilibrium occurs only in open vessel at constant temperature

27)

Find the active mass of 50g of $CaCO_3$ in 1 L vessel :-

(1) 0.5 mol/L
(2) 0.2 mol/L
(3) 1
(4) 0.25 mol/L

28) The rate contant for backward reaction is 7.5×10^{-4} and K_C is 1.5. Then calculate the value of rate constant for forward reaction :-

(1) 2×10^{-3}
(2) 5×10^{-4}
(3) 1.125×10^{-3}
(4) 9×10^{-4}

29) Ratio of active masses of 22 g CO_2, 3g H_2 and 7g N_2 in a gaseous mixture :-

(1) 22 : 3 : 7
(2) 0.5 : 3 : 7
(3) 1 : 3 : 1
(4) 2 : 6 : 1

30) Active mass of Pure solids & Pure liquids is assumed as one because :

(1) Only molecular weight remains constant
(2) Only density remains constant
(3) Both density and molar mass remains constant for pure solids & pure liquids
(4) None of the above

31) An equilibrium mixture of the reaction
$2H_2S_{(g)} \rightleftharpoons 2H_{2(g)} + S_{2(g)}$ has 5 mole H_2S, 1 mole H_2 and 4 mole S_2 in one litre vessel. The value of equilibrium constant (K_c) in mol litre^{-1} is :-

(1) 2

(2) 3

(3) $\frac{4}{25}$

(4) $\frac{8}{25}$

32)

The equilibrium constant (K_p) for the reaction is expressed as :

$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

(1) $K_p = (P_{NH_3})\ (P_{H_2S})$

(2) $K_p = \frac{(P_{NH_3})\,(P_{H_2S})}{(P_{NH_4HS})}$

(3) $K_p = (P_{NH_3})$

(4) $K_p = \frac{P_{NH_3}}{P_{H_2S}}$

33) For $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} + heat$:-
Which of the following relation is correct

(1) $K_p = K_c(RT)$

(2) $K_p = K_c\ (RT)^{-3}$

(3) $K_p = K_c\ (RT)^{-2}$

(4) $K_p = K_c(RT)^{-1}$

34) For the reaction
$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$ the units of K_p will be.

(1) atm

(2) $(atm)^3$

(3) $(atm)^{-2}$

(4) $(atm)^2$

35) For which reaction K_p is less than K_c at room temperature ?

(1) $N_2O_4\ (g) \rightleftharpoons 2NO_2(g)$

(2) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

(3) $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

(4) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

36) (a) The long form periodic table is based on increasing atomic weight.
(b) First d-block series starts with element scandium and ends with zinc.
(c) Alkali and alkaline earth metals are s-block elements.
(d) Chalcogens and halogens are p-block elements.

The correct statements are :-

(1) a, b and c
(2) a and b
(3) b, c and d
(4) all

37) Which group of elements follow triad rule?

(1) H, F, Cl
(2) Li, Na, K
(3) Cl, Br, I
(4) All

38) Atomic weight of an element X is 40 and that of element Z is 60. Atomic weight of their intermediate element Y, as per Dobereiner's triad will be :

(1) 40
(2) 50
(3) 60
(4) 100

39) If total 100 elements are present in periodic table then how many of them contain e^- in 4d subshell.

(1) 60
(2) 62
(3) 64
(4) 80

40)

Boron shows diagonal relationship with element having atomic number :-

(1) 10
(2) 12
(3) 14
(4) 13

41) Period number and group number of "Tantalum" (Z = 73) are respectively :

(1) 5, 7
(2) 6, 13
(3) 6, 5
(4) None of these

42) IUPAC name of element placed just after actinide series :

(1) Unniltrium (Unt)

(2) Unnilpentium (Unp)

(3) Ununbium (Uub)

(4) Unnilquadium (Unq)

43)

The outer electronic configuration of Gadolinium (atomic no. 64) is :

(1) [Xe] $4f^6\,5d^2\,6s^2$

(2) [Xe] $4f^7\,5d^1\,6s^2$

(3) [Xe] $4f^8\,6s^2$

(4) [Xe] $4f^5\,6s^2\,6p^3$

44) The element with the lowest atomic number that has a ground state electronic configuration of $(n-1)d^6\,ns^2$ is located in the :

(1) Fifth period

(2) Sixth period

(3) Fourth period

(4) Third period

45) In the fourth period, how many elements contain one or more 4d electrons ?

(1) 0

(2) 10

(3) 2

(4) 6

BIOLOGY

1) In basidiomycetes, karyogamy and meiosis takes place in :-

(1) Ascus

(2) Clamp connection

(3) Ascocarp

(4) Basidium

2) What is not true for Euglena :

(1) Presence of chlorophyll

(2) Presence of proteinaceous pellicle

(3) Presence of cellulosic cell wall

(4) Presence of Flagellum

3) *Nostoc* and *Anabaena* belong to :-

(1) Parasitic bacteria
(2) Archaebacteria
(3) Cyanobacteria
(4) Cocci bacteria

4) Which of the following group of bacteria could occur in gut of several ruminants ?

(1) Halophiles
(2) All archaebacteria
(3) Methanogens
(4) Thermoacidophiles

5) Match the following columns of fungi and their corresponding group :-

	Fungi		**Class**
(a)	*Alternaria*	(i)	Ascomycetes
(b)	Morels	(ii)	Phycomycetes
(c)	Smut fungi	(iii)	Deuteromycetes
(d)	Bread mould	(iv)	Basidiomycetes

(1) a-iii, b-i, c-ii, d-iv
(2) a-iii, b-i, c-iv, d-ii
(3) a-ii, b-i, c-iv, d-iii
(4) a-i, b-iii, c-iv, d-ii

6) Viruses that cause infections in plants mostly have :-

(1) ss RNA
(2) ds RNA
(3) ds DNA
(4) ss DNA

7) *Trypanosoma* is a :-

(1) Sporozoans
(2) Amoeboid protozoans
(3) Flagellated protozoans
(4) Ciliated protozoans

8) Mostly marine, photosynthetic and biflagellate organism are :-

(1) Diatoms
(2) Desmid
(3) Euglenoids

(4) Dinoflagellates

9) What is a bacteriophage ?

(1) A bacterium that feeds on viruses
(2) A virus that infects other viruses
(3) A virus that infects bacteria
(4) A bacterium that feeds on other bacteria

10) Viroids have

(1) ds DNA enclosed by protein coat
(2) ss DNA enclosed by protein coat
(3) ss RNA not enclosed by protein coat
(4) ds RNA enclosed by protein coat

11) Five kingdom system of classification is not based on :-

(1) Mode of nutrition
(2) Cell structure
(3) Body organisation
(4) Cell membrane

12) In Carl Woese's classification system, the domain that includes fungus-

(1) Bacteria
(2) Eukarya
(3) Monera
(4) Archaea

13) How many of the followings are photosynthetic. Dinoflagellates, Diatoms, Euglenoids, Slime mould, Protozoa :-

(1) Four
(2) Three
(3) Five
(4) Two

14) Characteristic feature of kingdom fungi is :

(1) It is a group of prokaryotic organisms
(2) It is a group of eukaryotic organisms with ingestive mode of nutrition
(3) It is a group of eukaryotic organisms with absorptive mode of nutrition.
(4) It is a group of eukaryotic autotrophic organisms

15) Which of the given organism(s) belong to kingdom protista according to five kingdom system?

(A) Virus (B) *Euglena* (C) Viroids (D) Lichen

(1) A and B

(2) C and D

(3) Only B

(4) A and C

16) Which of the following is/are common features of ascomycetes class ?
(I) Ascocarp
(II) Aseptate mycelium
(III) Conidiophores
(IV) Commonly found in aquatic habitats

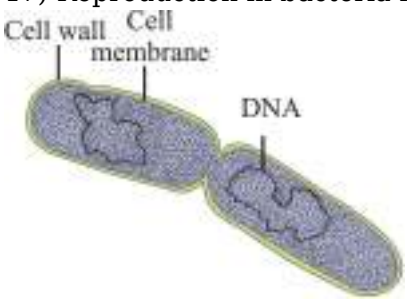
(1) I and III

(2) II, III and IV

(3) I, II and III

(4) II and III

17) Reproduction in bacteria is shown in following diagram. Select the correct match :-



	Mode of Reproduction	Conditions
(1)	Budding	Favourable
(2)	Binary fission	Favourable
(3)	Spore formation	Unfavourable
(4)	Budding	Unfavourable

(1) 1

(2) 2

(3) 3

(4) 4

18) Match the column-I with column-II and choose correct option :-

Column I		**Column II**	
(a)	Chrysophytes	(i)	Paramoecium
(b)	Dinoflagellates	(ii)	Euglena
(c)	Euglenoids	(iii)	Gonyaulax

(d)	Protozoans	(iv)	Diatoms

(1) a–i, b–iii, c–ii, d–iv

(2) a–ii, b–iv, c–iii, d–i

(3) a–iv, b–ii, c–iii, d–i

(4) a–iv, b–iii, c–ii, d–i

19) Identify the given figure and select the correct statements regarding it ?



(i) It is *Paramoecium*, having mixotrophic nutrition.
(ii) It is *Euglena* having one functional flagellum.
(iii) It is *Euglena* having pigments identical to higher plants.
(iv) It is *Paramoecium,* a ciliated protozoan.

(1) (i) & (iv)

(2) only (i)

(3) only (iii)

(4) (ii) & (iii)

20)

Which of the following statement is incorrect ?

(1) Mushroom belong to Ascomycetes

(2) Karyogamy in fungi occurs after plasmogamy

(3) Yeast are unicellular non-mycelial fungus

(4) Fungi bodies consist of long, slender thread like structures called hyphae.

21) Which of the following is true for the given figure A and B ?

(A)  (B) 

(1) Sexual reproduction in A occurs by ascospores

(2) Sexual reproduction in B occurs by zygospores

(3) Sexual reproduction in A and B occur by ascospores

(4) Sexual reproduction in B occurs by conidia

22) **Assertion :** Fungi show great diversity in morphology and habitat.
Reason : Sexual cycle in fungi involves three steps-
(i) Plasmogamy
(ii) Karyogamy

(iii) Meiosis

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

23) Which among the following spores are not involved in asexual reproduction of fungi?

(1) Conidia
(2) Zoospores
(3) Sporangiospores
(4) Ascospores

24) Which of the given statements is not true for viruses ?

(1) They are nucleoprotein where protein is infectious in nature
(2) They can be crystallised and consist largely of proteins
(3) Virus means venom or poisonous fluid
(4) Together a virus can never have both DNA & RNA as its genetic material

25) Prions are/have–

(1) DNA as genetic material
(2) RNA as genetic material
(3) Non infectious
(4) Abnormal folded protein

26) In Lichens, the algal component is called _____ and it is ____ while the fungal component is called _____ and it is _____.

(1) Phycobiont ; autotrophic; mycobiont; autotrophic
(2) Phycobiont; heterotrophic; mycobiont; autotrophic
(3) Mycobiont; heterotrophic; phycobiont; autotrophic
(4) Phycobiont; autotrophic; mycobiont; heterotrophic

27) Match the column I with column II and select the correct option from options given below:

	Column-I (Name of Fungi)		**Column-II (Class of Fungi)**
A	*Rhizopus*	P	Phycomycetes
B	*Neurospora*	Q	Ascomycetes
C	*Agaricus*	R	Deuteromycetes
D	*Trichoderma*	S	Basidiomycetes

(1) A - Q, B - P, C - S, D - R
(2) A - P, B - Q, C - R, D - S
(3) A - P, B - Q, C - S, D - R

(4) A - Q, B - P, C - R, D - S

28) **Assertion :** Lichens are very good pollution indicator.
Reason : Lichen do not grow in polluted area.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

29) Bacteriophages are made of both proteins and genetic material. Their genetic material is generally-

(1) ss DNA
(2) ds DNA
(3) ss RNA
(4) ds RNA

30) What is/are demerit(s) of two kingdom system of classification?

(1) It did not distinguish prokaryotes and eukaryotes
(2) It did not distinguish unicellular and multicellular organisms
(3) It did not separate photosynthetic and non-photosynthetic organisms
(4) All of the above

31) Which one of the following options gives the correct categorisation of A, B and C ?

	A Monera	**B** Protista	**C** Plantae
(1)	Blue green algae, *Mycoplasma*	Protozoans, Slime moulds	*Solanum, Mangifera*
(2)	Red algae, Archebacteria	Dinoflagellaes, Diatoms	*Mucor*, Yeast
(3)	Archebacteria, Eubacteria	Euglenoids, Green algae	Slime moulds, *Puccinia*
(4)	Eubacteria, Slime moulds	*Gonyaulax*, Diatoms	*Neurospora, Aspergillus*

(1) 1
(2) 2
(3) 3
(4) 4

32) Given table represents the characters of different kingdom according to Whittaker system of classification. Identify the kingdom A, B, C and D in given table on the basis of characters?

Characters	A	B	C	D
Cell Type	Eukaryotic	Prokaryotic	Eukaryotic	Eukaryotic
Cell Wall	Present	Present	Present in some	Absent
Body organisation	Loose tissue	Cellular	Cellular	Tissue/ organ/organ system
Mode of nutrition	Heterotrophic	Autotrophic and heterotrophic	Autotrophic and heterotrophic	Heterotrophic

(1) A: Monera; B: Protista; C: Plantae; D: Animalia

(2) A: Fungi; B: Monera; C: Plantae; D: Animalia

(3) A: Animalia; B: Monera; C: Plantae; D: Fungi

(4) A: Fungi; B: Monera; C: Protista; D: Animalia

33) **Statement-1:** Members of protista are eukaryotes.
Statement-2 : Protistan cell body does not contain a well defined nucleus and other membrane bounded cell organelles.

(1) Both statement 1 and statement 2 are correct

(2) Statement 1 is incorrect but statement 2 is correct

(3) Statement 1 is correct but statement 2 is incorrect

(4) Both statement 1 and statement 2 are incorrect

34) Match list I with list II and find the correct option:-

	List - I (Epithelial tissue)	List-II (Location)	
(A)	Cuboidal	1	Epidermis of skin
(B)	Ciliated	2	Inner lining of blood vessels
(C)	Brush-Border Columnar	3	Inner surface of gall bladder
(D)	Squamous	4	Inner lining of Fallopian tube
(E)	Keratinized squamous	5	Lining of pancreatic duct

(1) (A)–(5), (B)–(4), (C)–(2), (D)–(3), (E)–(1)

(2) (A)–(3), (B)–(4), (C)–(5), (D)–(2), (E)–(1)

(3) (A)–(5), (B)–(4), (C)–(3), (D)–(2), (E)–(1)

(4) (A)–(3), (B)–(4), (C)–(5), (D)–(1), (E)–(2)

35) Fill up the blanks by choosing correct combination of A, B, C and D :- 1. The ___A____is made up of a single thin layer of flattened cells with irregular boundaries.
2. ___B____ gland secrete digestive enzyme
3. ___C____ provide protection against chemical and mechanical stresses.
4. Cell junctions are found in ___D____ and other tissues.

(1) (A)-Cuboidal epithelium, (B)-Endocrine,
(C)-Columnar epithelium, (D)-Connective tissue

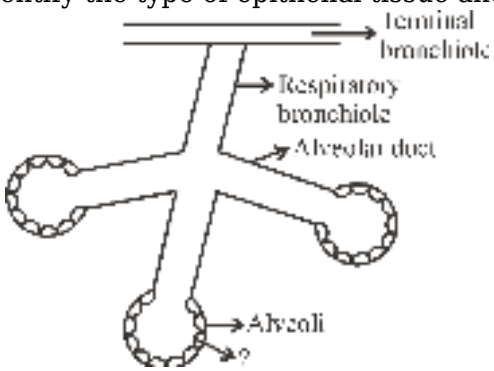
(2) (A)-Squamous epithelium, (B)-Exocrine,
(C)-Compound epithelium , (D)-Epithelial tissue

(3) (A)-Columnar epithelium, (B)-Liver,
(C)-Epithelial tissue, (D)-Muscular tissue

(4) (A)-Ciliated epithelium, (B)-Exocrine,
(C)-Connective tissue, (D)-Nervous tissue

36)

Identify the type of epithelial tissue and their function:-



(1) Simple cuboidal epithelium = absorption

(2) Simple columnar epithelium = secretion

(3) Simple squamous epithelium = diffusion

(4) Pseudostratified epithelium = filtration

37) In the given four statements, select the options which includes all the correct ones :-
(a) Secretory duct is present in endocrine glands and secretes hormones
(b) Mammary glands are exocrine glands
(c) Sweat gland is exocrine gland
(d) Bone has Nissl's granules
Option :-

(1) a, b, c

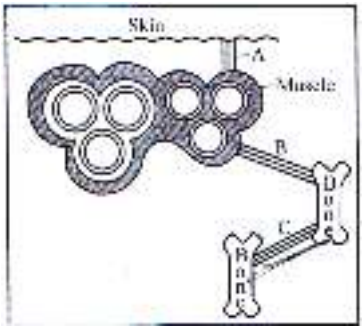
(2) b, c

(3) a, c, d

(4) b, d

38)

Identify the tissues A, B and C :



	A	B	C
(1)	Areolar connective tissue	Ligament	Tendon
(2)	Areolar connective tissue	Tendon	Tendon
(3)	Adipose connective tissue	Tendon	Ligament
(4)	Areolar connective tissue	Tendon	Ligament

(1) 1

(2) 2

(3) 3

(4) 4

39) Match the column and find out the correct answer.

	Column-I		Column-II
A	Most abundant formed elements of blood	(i)	Plateles
B	Collagen of areolar tissue is produced by	(ii)	Mast cells
C	Tendons are composed of this tissue	(iii)	Dense regular
D	Provide shape to external ear	(iv)	Elastic cartilage
		(v)	Hyaline cartilage
		(vi)	Erythrocytes
		(vii)	Fibroblast
		(viii)	Areolar tissue

(1) A-i B-viii C-iii D-v

(2) A-vi B-ii C-iii D-viii

(3) A-i B-vii C-viii D-iv

(4) A-vi B-vii C-iii D-iv

40) Match the column-I with II and select the correct answer :-

	Column-I		Column-II

(1)	Chondrocytes	a.	Bones
(2)	Mast cell	b.	Cartilage
(3)	Osteocytes	c.	Blood
(4)	Platelets	d.	Areolar tissue

(1) 1-d, 2-b, 3-a, 4-c

(2) 1-b, 2-d, 3-a, 4-c

(3) 1-b, 2-d, 3-c, 4-a

(4) 1-a, 2-b, 3-c, 4-d

41) What is incorrect for areolar tissue and adipose tissue?

(1) Loose connective tissue

(2) Present beneath the skin

(3) Both 1 and 2

(4) Dense connective tissue

42) Read the following four statements A-D : (A) Cuboidal epithelium is composed of a single layer of cube like cells. They are found in the wall of blood vessels and air sacs of lungs.
(B) The squamous epithelium is made up of a single layer of flattened cells with irregular boundries.
(C) Exocrine glands secrete mucus, saliva, earwax, oil milk and hormones.
(D) Simple epithelium cover the moist surface of the skin, the dry surface of bucal cavity, pharynx and inner lining of ducts of salivary glands.
How many of the above statements are not correct ?

(1) One

(2) Two

(3) Three

(4) All

43) Read the following statements and choose the correct option
Statement A: Exocrine glands secrete mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products.
Statement B: Some of the squamous or cuboidal cells get specialised for secretion and are called glandular epithelium.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

44) Blood is __A__ tissue which contain __B__ and __C__. Its main function is __D__.
Find out the correct match for A, B, C, D.

	A	**B**	**C**	**D**

1	Connective	RBC	WBC & Platelets	Conduction
2	Connective	WBC	Plasma	Lining & covering
3	Fluid connective	RBC	WBC & Platelets	Transportation
4	Epithelium	Plasma	RBC	Transportation

(1) 1

(2) 2

(3) 3

(4) 4

45) **Statement-I :** Inter cellular space is absent in Epithelial tissue.
Statement-II : Gap junction always transport big molecules and sometime small.

(1) Statement-I is right and Statement-II is wrong

(2) Statement-II is right and Statement-II is wrong

(3) Both Statement-I and Statement-II are wrong

(4) Both Statement-I and Statement-II are correct

46) All the listed glands pour their secretions into ducts except

(1) Salivary gland

(2) Digestive glands

(3) Pineal gland

(4) Mammary glands

47) Mast cells of connective tissue contain

(1) Vasopressin and relaxin

(2) Heparin and histamine

(3) Heparin and Calcitonin

(4) Serotonin and melanin

48) Which one is the wrong statement about the "bones" ?

(1) The bone-marrow in some bones is the site of production of blood-cells

(2) Bones have solid and pliable ground substance.

(3) Matrix of bones is rich in Ca^{+2} salts and collagen fibres.

(4) Bone cells are present in the spaces called lacunae.

49) Given below are two statement :- One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : In nearly all animal tissue, specialised junctions provide both structural and functional links between individual cells.

Reason (R) : Gap junctions help to stop substances from leaking across a tissue.
In the light of the above statements, choose the correct answer from the option given below:-

(1) (A) is not correct but (R) is correct

(2) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(3) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(4) (A) is correct but (R) is not correct

50) Which of the following statements are **incorrect** ?
(a) Each organ in our body is made of one or more type of tissues.
(b) Cell junction provide only structural links between it's individual cells.
(c) Tight junction help to stop substances from leaking across a tissue.
(d) Cuboidal epithelium is found in thin part of loop of Henle.
(e) All glands are composed of connective tissue

(1) a, b and c

(2) b, c and d

(3) b, d and e

(4) a, b, d and e

51) Read the following (A-D) statements :-
A - It is composed of single layer of tall and slender cells
B - Their nuclei are located at the base
C - Free surface may have microvilli
D - They are found in the walls of blood vessels and air sac of lungs
How many of the above statements are incorrect for simple columnar epithelium?

(1) Four

(2) Three

(3) Two

(4) One

52) Ciliated epithelial cells are known to occur in :-

(1) Eustachian tube and stomach

(2) Trachea and bronchioles

(3) Fallopian tube and ureters

(4) Trachea and oesophagus.

53) If a person has 8 liter of cardiac output & 75 ml of stroke volume then what will be the heart rate of human ?

(1) 100 Beat/min

(2) 106 Beat/min

(3) 95 Beat/min

(4) 75 Beat/min

54) In the process of blood coagulation, the conversion of prothrombin to thrombin requires which specific ion, and its absence would prevent clot formation even if all other factors are present?

(1) Sodium ions
(2) Potassium ions
(3) Calcium ions
(4) Magnesium ions

55) During Rh incompatibility, the condition erythroblastosis foetalis occurs due to which mechanism in subsequent pregnancies of an Rh-negative mother?

(1) Foetal Rh antigens crossing the placenta during first pregnancy.
(2) Maternal antibodies leaking into foetal blood and destroying RBCs.
(3) Direct mixing of maternal and foetal blood at birth.
(4) Foetal blood entering maternal circulation during delivery of second child.

56) The universal donor blood group 'O' can be transfused to all groups because it lacks which components on RBCs that could trigger agglutination?

(1) Antibodies in plasma
(2) Antigens on RBCs
(3) Clotting factors
(4) Rh factor

57) **Statement I:** The QRS complex in ECG represents atrial depolarisation.
Statement II: It initiates ventricular contraction.

(1) Both I and II are true
(2) I is true, II is false
(3) I is false, II is true
(4) Both I and II are false

58) Match the WBC types (Column I) with their percentages (Column II):

Column I		**Column II**	
(a)	Neutrophils	(i)	20-25%
(b)	Eosinophils	(ii)	0.5-1%
(c)	Basophils	(iii)	60-65%
(d)	Lymphocytes	(iv)	2-3%
(e)	Monocytes	(v)	6-8%

(1) a–ii, b–v, c–iv, d–iii, e–i
(2) a–v, b–iii, c–i, d–ii, e–iv
(3) a–iii, b–iv, c–ii, d–i, e–v
(4) a–iv, b–i, c–iii, d–v, e–ii

59) The hepatic portal system is unique because it carries blood from which organ to the liver before systemic circulation?

(1) Heart

(2) Lungs

(3) Intestine

(4) Kidneys

60) Arrange the blood vessel layers from inner to outer:
(i) Tunica externa
(ii) Tunica intima
(iii) Tunica media

(1) ii, iii, i

(2) i, iii, ii

(3) iii, ii, i

(4) ii, i, iii

61) **Statement I:** Cardiac output is stroke volume multiplied by heart rate.
Statement II: In a healthy individual, it averages 5 litres per minute.

(1) Both I and II are true

(2) I is true, II is false

(3) I is false, II is true

(4) Both I and II are false

62) **Assertion (A):** Angina occurs due to low oxygen to heart muscle.
Reason (R): Common in middle-aged and elderly.

(1) Both A and R are true, R explains A.

(2) Both A and R are true, R does not explain A.

(3) A is true, R is false

(4) A is false, R is true

63) Match the blood groups (Column I) with antibodies in plasma (Column II):

Column I	Column II
(a) A	(i) anti-A, anti-B
(b) B	(ii) nil
(c) AB	(iii) anti-A
(d) O	(iv) anti-B

(1) a–i, b–ii, c–iii, d–iv

(2) a–iii, b–iv, c–i, d–ii

(3) a–ii, b–i, c–iv, d–iii

(4) a–iv, b–iii, c–ii, d–i

64) In amphibians, the heart has three chambers with mixed blood in ventricle due to what?

(1) Complete double circulation
(2) Single ventricle
(3) Two ventricles
(4) Four chambers

65) Which of the following statements about red blood cells (RBCs) is incorrect?

(1) RBCs lack a nucleus in their mature form.
(2) RBCs contain hemoglobin for oxygen transport.
(3) RBCs have a lifespan of about 60 days.
(4) RBCs are produced in the bone marrow.

66) Which of the following statements about heart valves is incorrect?

(1) The bicuspid valve is located between the left atrium and left ventricle.
(2) The tricuspid valve prevents backflow into the right atrium.
(3) The pulmonary valve prevents backflow into the left ventricle.
(4) The aortic valve prevents backflow into the left ventricle.

67) Which of the following is correct about cardiac output?

(1) Cardiac output is the amount of blood pumped per beat.
(2) Cardiac output is calculated as heart rate × stroke volume.
(3) Cardiac output remains constant at all times.
(4) Exercise has no effect on cardiac output.

68) Which of the following blood components is involved in antigen-antibody reactions?

(1) Platelets
(2) Erythrocytes
(3) Globulins
(4) Fibrinogen

69) Which type of WBCs produces histamine during allergic reactions?

(1) Neutrophils
(2) Basophils
(3) Eosinophils
(4) Monocytes

70) Which of the following is responsible for ABO grouping?

(1) Presence or absence of clotting factors.

(2) Compatibility of blood groups during blood transfusion.

(3) Presence or absence of surface antigens (A and B) on WBCs.

(4) Presence or absence of two surface antigens (A and B) on the RBCs.

71) The first heart sound is

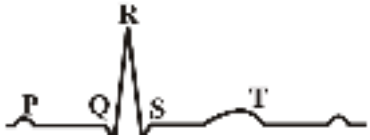
(1) 'LUBB' sound produced at the end of systole.

(2) 'DUP' sound produced at the end of systole.

(3) 'LUBB' sound produced at the beginning of systole.

(4) 'DUP' sound produced at the beginning of systole.

72) Given below is the ECG of a normal human. Which one of its components is correctly interpreted?



(1) Peak P and Peak R together - Systolic and diastolic blood pressures

(2) Peak P - Initiation of left atrial contraction only

(3) Complex QRS - One complete pulse

(4) Peak T - Initiation of total cardiac contraction

73) "X" is a fibrous tissue of the membranous septum of the heart just above the septal cusp of the tricuspid valve. It separates the atrium and the ventricle of the same side. Identify "X".

(1) Sino atrial node

(2) Atrioventricular septum

(3) Atrioventricular node

(4) Interventricular septum

74) Match the blood vessels of human listed under column I with the functions given under column II. choose the answer which gives the correct combination of the alphabets of the two columns.

	Column I		**Column II**
A.	Renal portal	p.	carries blood to almost whole of the body
B.	Pulmonary vein	q.	is a portal vein
C.	Dorsal aorta	r.	Carries oxygenated blood
D.	Pulmonary artery	s.	Carries deoxygenated blood
		t.	Absent in man

(1) A = t, B = p, C = r, D = q

(2) A = t, B = r, C = p, D = s

(3) A = s, B = t, C = r, D = p

(4) A = t, B = p, C = q, D = r

75) Which of the following substance, if introduced in blood stream would cause coagulation of blood at the site of its introduction ?

(1) Prothrombin

(2) Fibrinogen

(3) Thromboplastin

(4) Heparin

76) Choose the correct statements is given below options.
I. Artery and veins consists of 3 layers.
II. Left ventricle pumps into the aorta.
III. Normal activity of human heart are regulated intrinsically i.e. auto-regulated.
IV. Tunica media is comparatively thick in vein.

(1) I, II, III

(2) I and II only

(3) I and III only

(4) I, II, III, IV

77) When all four chambers of heart are in relaxed in state that is –

(1) Systolic mode

(2) Diastolic and Systolic mode respectively

(3) Joint Diastole

(4) Contraction mode

78) Nodal tissue which is present in the right upper corner of the right Atrium is –

(1) Sino-atrial node

(2) Atrio-Ventricular node

(3) AV bundle

(4) Purkinjee fibre

79) Which one of the following plasma proteins is involved in the coagulation of blood?

(1) Serum amylase

(2) A globulin

(3) Fibrinogen

(4) An albumin

80)

	Column-I		**Column-II**
A.	Erythrocytes	i.	1,50,000-3,50,000/mm^3

B.	Thrombocytes	ii.	5-5.5 million/mm^3
C.	Leucocytes	iii.	6000-8000/mm^3

Options :

(1) A-i, B-ii, C-iii

(2) A-iii, B-i, C-ii

(3) A-ii, B-iii, C-i

(4) A-ii, B-i, C-iii

81)

A person with type A Rh– blood can receive a blood transfusion from which of the following types.
(i) A Rh^+ (ii) B Rh^- (iii) AB Rh^- (iv) O Rh^- (v) A Rh^-

(1) i and iii

(2) iii and v

(3) iv and v

(4) i, iv and v

82) Match column-I with column-II.

	Column-I		**Column-II**
(A)	Cardiac arrest	(i)	Heart not pumping blood effectively
(B)	Heart failure	(ii)	Heart muscle is suddenly damaged
(C)	Heart attack	(iii)	Acute chest pain
(D)	Angina	(iv)	Heart stop beating

(1) A-i, B-ii, C-iii, D-iv

(2) A-iv, B-ii, C-i, D-iii

(3) A-iv, B-i, C-ii, D-iii

(4) A-ii, B-iii, C-i, D-iv

83)

The correct route through which impulse travels in the heart is

(1) SA node → Purkinje fibres → Bundle of His → AV node → Heart muscles

(2) SA node → AV node → Bundle of His → Purkinje fibres → Heart muscles

(3) AV node → Bundle of His → SA node → Purkinje fibres → Heart muscles

(4) AV node → SA node → Purkinje fibres → Bundle of His → Heart muscles

84) Human heart is derived from

(1) Ectoderm

(2) Mesoderm

(3) Endoderm

(4) Both (1) & (3)

85) How much protein present in plasma ?

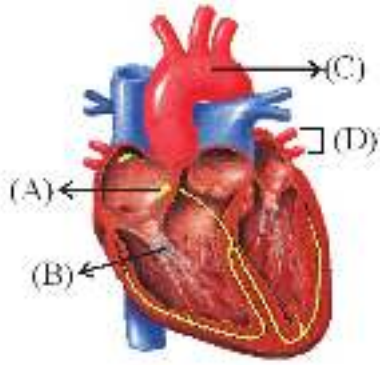
(1) 6-8 %

(2) 8-10 %

(3) 45 %

(4) 2-3 %

86) Identify the labels and their function.



	Labelled structures	Functions
(1)	A = SAN	Generates 70-75 impulses/min
(2)	B = Purkinje fibres	Conducts impulses to entire ventricles
(3)	C = Aorta	Carries oxygenated blood to tissues/organs
(4)	D = Pulmonary vein	Carries deoxygenated blood to lungs

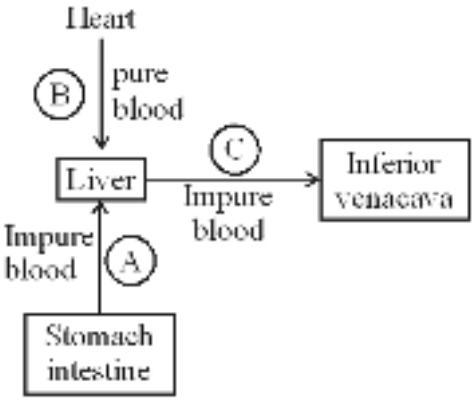
(1) 1

(2) 2

(3) 3

(4) 4

87)



	A	B	C
(1)	Hepatic vein	Hepatic portal vein	Hepatic artery
(2)	Hepatic portal vein	Hepatic artery	Hepatic vein
(3)	Hepatic artery	Hepatic vein	Hepatic portal vein
(4)	Hepatic portal vein	Hepatic vein	Hepatic artery

(1) 1

(2) 2

(3) 3

(4) 4

88) Which of the following lists of the valves in the order through which the blood flows from the venacava through the heart ?

(1) Tricuspid→Pulmonary semilunar→bicuspid→aortic semilunar

(2) Mitral→pulmonary semilunar→bicuspid→aortic semilunar

(3) Aortic semilunar→pulmonary semilunar→tricuspid→bicuspid

(4) Bicuspid→aortic semilunar→tricuspid→pulmonary semilunar

89) Fill in the blanks
The first heart sound (lub) is associated with the __A__ whereas the second heart sound (dub) is associated with __B__.

	(A)	(B)
(1)	Closure of AV valves	Closure of semilunar valves
(2)	Closure of bicuspid valve	Closure of tricuspid valve
(3)	Opening of bicuspid and tricuspid valves	Closure of semilunar valves
(4)	Opening of AV valves	Opening of semilunar valves

(1) 1
(2) 2
(3) 3
(4) 4

90) During ventricular systole :-

(1) Semilunar valves are closed
(2) About 30 percent blood is pumped in arteries
(3) Tricuspid and bicuspid valves are closed
(4) Ventricular pressure declines

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	2	1	1	4	3	4	2	4	3	4	3	3	1	4	2	3	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	4	3	4	2	2	4	1	2	3	2	1	4	2	2	2	3	2	3
Q.	41	42	43	44	45															
A.	3	3	3	2	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	4	4	2	3	3	4	3	2	2	2	2	1	2	2	2	4	2	1	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	1	1	2	4	4	3	3	4	3	3	1	3	4	3	3	4	2	2	3
Q.	86	87	88	89	90															
A.	3	4	2	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	3	3	3	2	1	3	4	3	3	4	2	2	3	3	1	2	4	4	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	2	4	1	4	4	3	1	2	4	1	4	3	3	2	3	2	4	4	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	3	3	3	3	2	2	4	3	4	2	2	3	2	2	3	3	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	2	4	2	3	3	2	3	2	4	3	3	2	2	3	1	3	1	3	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	3	2	2	1	3	2	1	1	3										

SOLUTIONS

PHYSICS

1) $\vec{r} = 7\hat{i} + 3\hat{j} + \hat{k}$

$\vec{F} = -3\hat{i} + \hat{j} + 5\hat{k}$

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 3 & 1 \\ -3 & 1 & 5 \end{vmatrix}$$

$\vec{\tau} = 14\hat{i} - 38\hat{j} + 16\hat{k}$

2)

Projection of $\vec{a}$ on $\vec{b}$

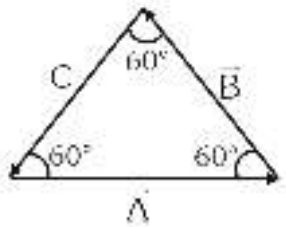
$$\frac{\vec{a}.\vec{b}}{|\vec{b}|} = \frac{5}{\sqrt{6}}$$

3) $\left(\vec{a} + 2\vec{b}\right) . \left(5\vec{a} - 4\vec{b}\right) = 0$

$5a^2 - 8b^2 - 4\vec{a}.\vec{b} + 10\vec{a}.\vec{b} = 0$

$6\vec{a}.\vec{b} = 3 \Rightarrow \vec{a}.\vec{b} = \frac{1}{2} \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$

4) So, angle between $\vec{A}$ and $\vec{B}$, $\vec{B}$ and $\vec{C}$, $\vec{C}$ and $\vec{A}$ is 120° each.



Therefore,

$\vec{A} \cdot \vec{B} + \vec{B} \cdot \vec{C}$

$= 1 \times 1 \times \cos 120^\circ + 1 \times 1 \times \cos 120^\circ$

$= -\frac{1}{2} - \frac{1}{2} = -1$

5) ABcosθ = $2\sqrt{3}$ .....(i)

ABsinθ = 2 .....(ii)

(ii) ÷ (i)

$\tan\theta = \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$

θ = 30º

6) Cross product of two scalar quantities is not meaningful.

7) **Question Explanation :** Two parallel vectors is given, calculate value of m × n.

Concept: Parallel vectors.

Solution:
$\vec{a} = 2\hat{i} + 6\hat{j} + m\hat{k}$
$\vec{b} = n\hat{i} + 12\hat{j} + 4\hat{k}$
For parallel vectors

$$\frac{2}{n} = \frac{6}{12} = \frac{m}{4}$$

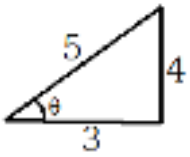
n = 4 m = 2, ⇒ mn = 8

Final answer: option (3)

8) **Explain question:**
To find vector perpendicular to ($a\cos\theta\hat{i} + b\sin\theta\hat{j}$)

Concept:
Dot product.

Solution:
If two vector are perpendicular then their dot product will be zero. (1)
$\left(a\cos\theta\hat{i} + b\sin\theta\hat{j}\right) . \left(b\sin\theta\hat{i} - a\cos\theta\hat{j}\right) = 0$
$ab\sin\theta\cos\theta - ab\sin\theta\cos\theta = 0$
(2) $\left(\frac{\sin\theta\hat{i}}{a} - \frac{\cos\theta\hat{j}}{b}\right) . \left(a\cos\theta\hat{i} + b\sin\theta\hat{j}\right) = 0$
$\sin\theta\cos\theta - \sin\theta\cos\theta = 0$
(3) $\left(a\cos\theta\hat{i} + b\sin\theta\hat{j}\right) .5a\hat{k} = 0$
Final answer: option (4)

9) 

$A \times B = AB \sin\theta = 4$

$A \cdot B = AB \cos\theta = 3$

$\tan\theta = \frac{4}{3}$

$\cos\theta = \frac{3}{5}$

$R = \sqrt{A^2 + B^2 + 2AB \times \frac{3}{5}}$

10) **Explain Question:**
To find work done

Concept:
Application of dot product

Solution:
displacement $\vec{s} = \vec{r}_2 - \vec{r}_1$

$\vec{s} = \left(4\hat{i} - 7\hat{j} + 10\hat{k}\right)$ m

$W = \vec{F}.\vec{S}$

$= \left(10\hat{i} - 3\hat{j} + 6\hat{k}\right) . \left(4\hat{i} - 7\hat{j} + 10\hat{k}\right)$

W = (40 + 21 + 60) = 121 Joule

Final answer: Option (4)

11) $\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 4 \\ 1 & 4 & -5 \end{vmatrix}$

$= \hat{i}(15 - 16) - \hat{j}(-10 - 4) + \hat{k}(8 + 3)$

$= -\hat{i} + 14\hat{j} + 11\hat{k}$

12)

Area of parallelogram = $\frac{1}{2}|\vec{P} \times \vec{Q}|$

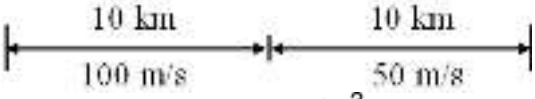
where, $\vec{P} = 2\hat{i} + 3\hat{j}, \vec{Q} = \hat{i} + 4\hat{j}$

 Area = 2.5 units.

13) $|\vec{a}_{AV}| = \frac{|\Delta\vec{v}|}{\Delta t} = \frac{|40\hat{i} - 30\hat{j}|}{20}$

$= \frac{50}{20} = 2.5 \text{ km/sec}$

14) **To find:** The average speed over the whole distance.
Concept : Speed and velocity (average and instantaneous)
Solution :



Average speed $= \frac{20 \times 10^3}{\frac{10^4}{100} + \frac{10^4}{50}}$

$= \frac{2}{\frac{1}{100} + \frac{1}{50}}$

$= \frac{2}{\frac{1}{50}\left(\frac{1}{2} + 1\right)}$

$= \frac{100}{3/2}$

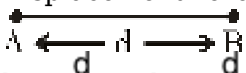
$= \frac{200}{3} = 66.67$ m/s

Final answer :
Option (3) is correct.

15) **Explaining :** find average speed & average velocity for whole journey
Concept : avg. speed $= \frac{\text{total distance}}{\text{total time}}$

avg. velocity $= \frac{\text{net displacement}}{\text{total line}}$

Solution :
Displacement zero ⇒ average velocity = 0



$t_1 = \frac{d}{40}$; $t_2 = \frac{d}{30}$

Avg. speed $= \frac{2d}{\frac{d}{40} + \frac{d}{30}} = \frac{2400}{70}$

$= 34.3 \text{km/hr}$

Final Answer : (1)

16)

$S = t^3 - 6t^2 + 3t + 4$
$v = 3t^2 - 12t + 3$
$a = 6t - 12$
$a = 0$ at $t = 2$ sec
$v)_{t=2} = -9$ m/s

17) **Explain question :** Velocity depends on the position, then acceleration at the given position will be.
Concept: This question is based on motion with variable acceleration.
Solution :
$v = 3x^2 - 2x$
$\frac{vdv}{dx} = a = \left(3x^2 - 2x\right)(6x - 2)$
$a(x = 2) = (3(2)^2 - 2 \times 2)(6 \times 2 - 2)$
$= (3 \times 4 - 4)(12 - 2)$
$= 8 \times 10 = 80\ m/s^2$
Final answer : $80\ m/s^2$

18)

From Galileo law of odd numbers, if particle starts from rest and moves with constant acceleration then in successive equal interval of time the ratio of distance covered by it will be 1:3:5:7:.....(2n-1)
$S_1 : S_2 : S_3 = 1 : 3 : 5$
$S_1 = \frac{S_2}{3} = \frac{S_3}{5}$

19)

Correct Answer: (3)

20)

$100 = -40\ t + \frac{1}{2} \times 10 \times t^2$
$\Rightarrow t^2 - 8t - 20 = 0$
$\Rightarrow t = -2, 10$
Since 't' cannot be negative, t = 10 s is answer.

21) **Question Explanation :** To find the maximum height attained by the rocket from ground when it is projected upward.
Concept : Analysis of Velocity-time graph.
Solution : Displacement (height) = Area under velocity-time graph (area till the point at which v = 0 since at maximum height, v = 0)
$= \frac{1}{2} \times 100 \times 600$
$= 30000$
$= 30\ km$
Final Answer : Option (4)

22) $h = 0 + \frac{1}{2}gt_1^2$
$\Rightarrow\ t_1 = \sqrt{\frac{2h}{g}} t_1 = \sqrt{\frac{2h}{g}}$

$$\frac{h}{2} = 0 + \frac{1}{2}gt_2^2 \Rightarrow t_2^2 = \frac{2}{g} \times \frac{h}{2}$$
$$\Rightarrow t_2 = \sqrt{\frac{2}{g} \times \frac{h}{2}} = \frac{1}{\sqrt{2}} \times \sqrt{\frac{2h}{g}} = \frac{t_1}{\sqrt{2}}$$

23) **Question Explanation :** To find the time for a ball to strike the ground which is thrown upwards from the tower of height 40m with 10 m/s.
Concept : This question is based on motion under gravity.
Solution :
$$S = ut + \frac{1}{2}at^2$$
Putting the values with sign convention (upwards → +ve, downwards → –ve)
$$-40 = 10(t) - \frac{1}{2} \times g \times t^2$$
$$-40 = 10t - 5t^2$$
$$t^2 - 2t - 8 = 0$$
$$(t - 4)(t + 2) = 0$$
t = 4 sec t = –2 is not possible
Final Answer : Option (4)

24) **Explain question :** In this question, a body is dropped from a tower. In the last second it covers a distance 9x where 'x' is the distance covered in the first second. We need to find the total time of fall.
Concept : This question is based on motion under gravity (free fall)
Solution :
Distance covered in first second,
$$x = \frac{1}{2} \times g \times t^2 = \frac{1}{2} \times 10 \times 1^2 = 5 \text{ unit}$$
if total time of journey is 'n' sec (say), then distance travelled in last second (n^{th} sec)
$$9x = \frac{1}{2} \times g \times (2n - 1)$$
$$\Rightarrow 9 \times 5 = \frac{1}{2} \times 10 \times (2n - 1)$$
⇒ n = 5 seconds
Final answer : option 3

25) Average speed = $\frac{20}{10} = 2 \text{ ms}^{-1}$
Average velocity = $\frac{0}{10} = 0 \text{ ms}^{-1}$

26)

Average velocity = $\frac{\text{displacement}}{\text{time interval}}$
$= \frac{-5-5}{5} = -2$

27) Average speed = $\frac{\text{distance}}{\text{time}} = \frac{3+3+3}{3}$ = 3 m/s

28)

Explain the Question : Given here, Velocity ⇒ v = 2t(3 – t)

= $(6t - 2t^2)$ m/s

At what time "t" the velocity 'v' is maximum = ?

Concept : This question is based on application of calculus to find the condition to find the maximum velocity.

Explanation : Velocity ⇒ v = $(6t - 2t^2)$

Now, $\frac{dv}{dt} = \frac{d}{dt}(6t - 2t^2)$

= $(6 - 4t)$ m/s^2

Putting $\frac{dv}{dt} = 0$

⇒ 6 – 4t = 0

⇒ 6 = 4t

$\Rightarrow t = \frac{3}{2}$ sec &

$\frac{d^2v}{dt^2} = \frac{d}{dt}\left(\frac{dv}{dt}\right)$

$= \frac{d}{dt}(6-4t)$

= 0 – 4

= – 4 m/s^3

At $t = \frac{3}{2} \Rightarrow \frac{d^2v}{dt^2} = -4 < 0$ (negative)

so, $t = \frac{3}{2}$ is the point of maxima

Hence, the velocity will be maximum at t = $\frac{3}{2}$ sec

Final Answer : 4

29) **Question Explanation :** To find the average velocity for t = 0 to t = 40 sec (time interval).

Concept : Speed and velocity (Average and instantaneous)

Solution :

$V_{avg} = \frac{\text{Area under curve}}{\text{time}}$

$= \frac{\frac{1}{2} \times 20 \times 20 + 10 \times 20 + \frac{1}{2} \times (20+60) \times 10}{40}$

= 20 m/s

Final Answer : Option (1)

30) **Explanation Question:**

The velocity of a particle is given as a function of its position, v=$\sqrt{x+2}$. find the acceleration of the particle at the specific position x=2 m

Concept:

This question is based on kinematics - calculus

Formula :

$a = V\frac{dv}{dx}$

Solution:

$V = \sqrt{x+2}$

$a = V\frac{dv}{dx} = \sqrt{x+2}\frac{d}{dx}\sqrt{x+2}$

$a = \sqrt{x+2} \times \frac{1}{2(\sqrt{x+2})} \times 1$

$a = 0.5\ m/sec^2$

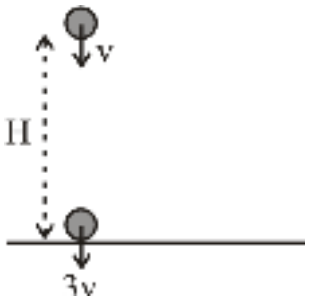
Final Answer: (2)

31)

Question Explanation : To find the Height of the tower if a ball thrown from it with velocity 'v' reaches the ground with '3v' speed

Concept : This question is based on motion under gravity

Solution :



By the equation

$v^2 = u^2 + 2aS$

$(3v)^2 = v^2 + 2gH \quad (a = g)$

$H = \frac{4v^2}{g}$

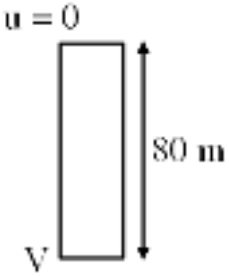
Final answer : option (3)

32)

Question Explanation : To find the velocity of a stone just before striking the ground which is dropped from 80 m height.

Concept : This question is based on motion under gravity / free fall.

Solution :



$V^2 = u^2 + 2aS$

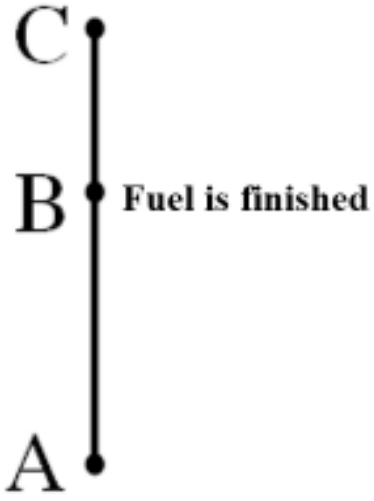
$V^2 = 0 + 2 \times 10 \times (80) \qquad (a = g = 10\ m/s^2)$

$V^2 = 20 \times 80$

$V = \sqrt{1600}$
$V = 40$ m/s
Final answer : option (2)

33)

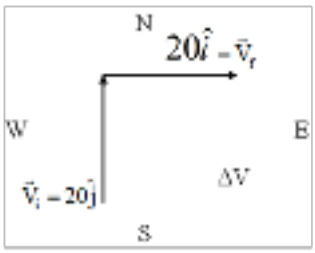
Question Explanation : In this question, we need to find the time from the instant of firing, at which the rocket will attain the maximum height.
Concept : This question is based on motion under gravity.
Solution :
Let the rocket be fired from point A
For A to B



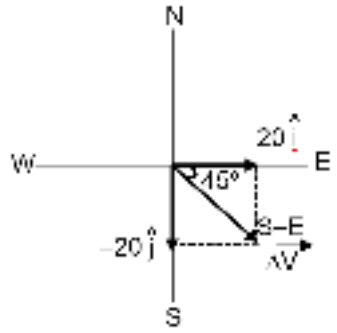
$V_B = u_A + at$
$V_B = 0 + 20 \times 30$ ($a = 20$ m/s^2)
$V_B = 600$ ms^{-1}
For B to C
by the equation, $v = u + at$
$V_C = V_B - gt$ ($a = -g$)
$0 = 600 - 10t$ (c is the point of maximum height attained)
$t = 60$ sec
total time to attain max height = 60 + 30 = 90 sec
Final Answer : Option (1)

34) T. Distance = $u\, t_r + \frac{u^2}{2a}$
$= 15 \times 0.4 + \frac{15 \times 15}{2 \times 5}$
$= 6 + 22.5 = 28.5$ m

35) **Question Explanation :** To find the change in velocity of truck.
Concept : This question is based on change in velocity.
Formula : $\Delta\vec{V} = \vec{V}_f - \vec{V}_i$
Solution :



$\vec{V}_f = 20\hat{i}$
$\vec{V}_i = 20\hat{j}$
$\Delta\vec{V} = \vec{V}_f - \vec{V}_i$
$\Delta\vec{V} = 20\hat{i} - 20\hat{j}$
$\left|\Delta\vec{V}\right| = \sqrt{400 + 400}$
$= 20\sqrt{2}$
$\Delta\vec{V} = 20\hat{i} - 20\hat{j}$



$\Delta\vec{V} = 20\sqrt{2}$ m/s S-E
Final Answer : Option (2)

36) $x = 5t - 2t^2$ $y = 10\ t$

$\frac{dx}{dt} = 5 - 4t$ $\frac{dy}{dt} = 10$

$\frac{d^2x}{dt^2} = -4$ $\frac{d^2y}{dt^2} = 0$

So –4 m/s^2

So, acceleration $\Rightarrow a = a_x\hat{i} + a_y\hat{j}$

37) Acceleration is non-zero for t = 20 to t = 40s
So, area covered = Displacement (S)

$\Rightarrow \quad S = \frac{1}{2}.20.3 + 1 \times 20 = 50$

38)

$a = (2x + 1)$

$V\frac{dv}{dx} = (2x + 1)$

$$\int_4^V V\,dv = \int_0^3 (2x+1)\,dx$$

$$\left(\frac{V^2}{2}\right)_4^V = \left(\frac{2x^2}{2} + x\right)$$

$$\frac{V^2 - 16}{2} = (9+3)$$

$V^2 = 24 + 16$

$V = \sqrt{40}$ m/sec

39) $\langle v \rangle = \frac{2v_1v_2}{v_1+v_2} = \frac{2v_0 5v_0}{6v_0} = \frac{5v_0}{3}$

40)

From given graph

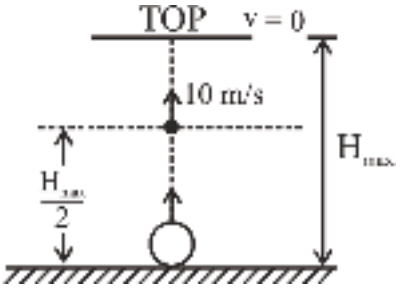
$V = V_0 - \frac{V_0}{x_0}x$

$a = V\frac{dV}{dx} = \left(V_0 - \frac{V_0}{x_0}x\right)\left(-\frac{V_0}{x_0}\right)$

$a = \left(\frac{V_0}{x_0}\right)^2 x - \frac{V_0^2}{x_0}$

this is equation of straight line of positive slope and negative intercept so option (3) is correct

41)



From, $v^2 = u^2 - 2gh$

for half of H_{max} to H_{max}

$0 = (10)^2 - 2g\left(\frac{H_{max}}{2}\right)$

or $H_{max} = 10$ m

42) Let the total time of journey be n seconds.

Using, $S_n = U + \frac{a}{2}(2n-1)$

$\Rightarrow 45 = 0 + \frac{10}{2}(2n-1) \Rightarrow n = 5$ sec .

height of tower ; $h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 5^2 = 125m$

43) As Given $S_{n^{th}} = S_5$ [Also u = 0 & a = g]

$\Rightarrow \frac{g}{2}(2n-1) = \frac{1}{2}g(5)^2$

$\Rightarrow$ n = 13 sec

44)

$H = \frac{1}{2}gt^2$

$h = \frac{1}{2}g\left(\frac{t}{4}\right)^2 = \frac{H}{16}$ from releasing point

so distance from the ground

= H – H/16 = 15H/16

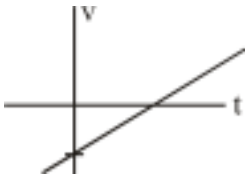
45) As $x = 8 - 4t + 6t^2$

$\Rightarrow v = \frac{dx}{dt} = 0 - 4 + 12t$ or $v = 12t - 4$

As t = 0, v = –4. Also $v \propto t$

Hence, (1) represents the correct graph.

OR



$x = 8 - 4t + 6t^2$

$v = -4 + 12t$

CHEMISTRY

46)

Relation between erg and Joule –

$1J = 10^7$ erg

In option (3) :-

$1 \text{ erg} = 10^7$ J (Incorrect)

Final Answer: option (3)

47)

No. of moles $= \frac{\text{mass}}{\text{molar mass}} = \frac{90}{18} = 5\text{moles}$

48) $O_3 = \frac{48}{48} = 1 \times N_A$

$CO_2 = \frac{44}{44} = 1 \times N_A$

$SO_2 = \frac{64}{64} = 1 \times N_A$

$H_2 = \frac{8}{2} = 4 \times N_A$ (Max. no. of molecules)

49) $n = \frac{W(g)}{Mw} = \frac{V(L)}{22.4}$

$\frac{1.25}{Mw} = \frac{1}{22.4}$

Mw = 1.25 × 22.4 = 28 g/mol

50) No. of atoms = (Mole × N_A) atomicity
= (1 × N_A) × 7
= 7 N_A

51)

Molecular mass of $CaCO_3$ = 100 amu

$\because 1 amu = 1.67 \times 10^{-24} g$

$100 amu = 1.67 \times 10^{-22} g$

52)

Molecular weight = vapour density × 2 = 128 × 2 = 256 g
⇒ 64 × x = 256
⇒ x = 4

53)

C	H	O
8	12	4
2	3	1

C_2H_3O

54)

Atomicity = 1

$11.5 = \frac{1 \times 23}{\text{Min. molecular mass}} \times 100$

Min. mm = 200 amu

55) $S = \frac{40}{32} = 1.25 \Rightarrow \frac{1.25}{1.25} \Rightarrow 1$
$O = \frac{60}{16} = 3.75 \Rightarrow \frac{3.75}{1.25} \Rightarrow 3$
$\therefore E.F = SO_3$

56)

Number of Cr – atoms = 4.8×10^{10}
Number of O – atoms = 9.6×10^{10}
Simplest ratio ⇒ Cr : O ⇒ 1:2
simplest whole Number ration is 1 Cr and
2 O – atoms,
Therefore empirical formula is CrO_2
Final Answer: Option (2)

57) **Explain Question:** We are given vapour density (VD) of a gas = 50. We need to find its molecular weight.
Concept: This question is based on Vapour Density (VD)=Molecular Weight/2
Solution:
Molecular Weight = 2 × VD
Molecular Weight = 2 × 50 = 100
Final Answer: Option 2

58)

$2Mg + O_2 \rightarrow 2MgO$
1 mole $O_2 \rightarrow$ 2 mole Mg
1.5 mole $O_2 \rightarrow 1.5 \times 2$ mole Mg
= 3 mol Mg

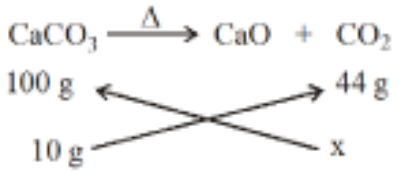
59)

No. of atoms = Atomicity × mole × N_A
(A) No. of atoms = $6 \times \frac{212}{106} \times N_A = 12N_A$
(B) No. of atoms = $3 \times \frac{248}{62} \times N_A = 12N_A$
(C) No. of atoms = $3 \times \frac{240}{40} \times N_A = 18N_A$
(D) No. of atoms = $2 \times \frac{12}{2} \times N_A = 12N_A$
(E) No. of atoms = $3 \times \frac{220}{44} \times N_A = 15N_A$

60) **Molecular mass of butane:**
= (4 × 12) + (10 × 1) = 48 + 10 = 58 amu

Molar mass of butane:
= 58 g/mol
So, 1 mole of butane contains 6.022×10^{23} molecules and weighs 58 g
Final Answer: (2) Statement-I and statement-II both are correct

61)



$\Rightarrow$ 100 x = 44 × 10 and x = 4.4 g

$= \frac{4.4}{44} \times 22.4$ = 2.24 L of CO_2 at STP

62)

Initial mole = $\frac{120}{100} = 1.2\,\text{mol}$

removed moles = $\frac{6 \times 10^{22}}{6 \times 10^{23}} = 0.1\,\text{mol}$

remaining moles = 1.2 – 0.1
= 1.1 mole

63) $C_2H_6(g) + \frac{7}{2}O_2 \rightarrow 2CO_2(g) + 3H_2O(g)$

$n_{C_2H_6} = \frac{450}{30} = 15\,\text{mol}$

$n_{O_2} = \frac{7/2}{1} \times 15 = 52.5\,\text{mol}$

$W_{O_2} = 52.5 \times 32 = 1680\,\text{g} = 1.68\,\text{kg}$

64)

1. 1g atom of C :
1g atom of C = 1 mole of C
Molar mass of C = 12 g/mol
Mass = 1 mol × 12 g/mol = 12g
2. 1/2 mole of CH_4 :
Molar mass of CH_4 = 12 + (4 × 1) = 16 g/mol
Mass = 0.5 mol × 16 g/mol = 8g
3. 10 mL of water :
Density of water ≈ 1g/mL
Mass = 10 mL × 1 g/mL = 10g
4. 3.011×10^{23} atoms of oxygen :
Moles of O atoms = (3.011×10^{23}) (6.022×10^{23}) = 0.5 mol
mass of O = 0.5 mol × 16 g/mol = 8g
the highest mass 1g atom of C (12 lag) option 1
Final answer : 1

65)

$$\begin{array}{ccccc} & 2C & + & O_2 \rightarrow & 2CO \\ \text{moles} & \frac{24}{12} & & \frac{96}{32} & \\ & = 2 \text{ mol} & & = 3 \text{ mol} & \end{array}$$

$V_1 = \frac{2}{2} = 1 \qquad V_2 = \frac{3}{1} = 3$

$V_1 < V_2$

$\Rightarrow$ C is L.R.

66)

N^{3-} has 10 electrons.
Weight of N^{3-} = 14g
Molar mass of N = 14 g/mol
Moles of N^{3-} = 14/14 = 1mol.
Number of ions = $1 \times N_A = N_A$.
Total electrons = $N_A \times 10 = 6.023 \times 10^{24}$
Final Answer : Option (3)

67) $n_{O_2} = \frac{8}{32} = 0.25$ mole

$n_{H_2} = \frac{4}{2} = 2\text{mole}$

$n_{He} = \frac{52}{4} = 13\text{mole}$

$n_{Cl_2} = \frac{35.5}{71} = 0.5$ mole

68) $n_{solute} = \frac{5.85}{58.5} = \frac{1}{10}$

$Vsol^n = \frac{20}{1000} = \frac{1}{50}L$

$M = \frac{n_{solute}}{V_{sol^n}(L)} = \frac{1/10}{1/50} = \frac{50}{10} = 5\ M$

69)

Pure mass = $200 \times \frac{50}{100} = 100$ g

$n_{CaCO_3} = \frac{100}{100} = 1$ mol

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

$n_{CO_2} = \frac{1}{1} \times 1 = 1$ mol

$V_{CO_2} = 1 \times 22.4 = 22.4\,L$

70)

Combustion reaction is an exothermic reaction.

71)

Equilibrium occurs only in open vessel at constant temperature is not characteristics of equilibrium.

72)

The active mass of a pure solid (like $CaCO_3$) is taken as unity (1), regardless of the amount present.

73) $K_C = \frac{K_f}{K_b}$

$1.5 = \frac{K_f}{7.5 \times 10^{-4}}$

$K_f = 11.25 \times 10^{-4}$

$= 1.125 \times 10^{-3}$

74)

Ratio of active mass

$P_{CO_2} : P_{H_2} : P_{N_2} = n_{CO_2} : n_{H_2} : n_{N_2}$

$= \frac{22}{44} : \frac{3}{2} : \frac{7}{28}$

$= \left(\frac{1}{2} : \frac{3}{2} : \frac{1}{4}\right) \times 4$

$= 2 : 6 : 1$

75)

Active mass as concentration

$[\ ] = \frac{\text{mole}}{\text{vol.}} = \frac{W}{Mw \times \text{vol.}}$

$[\ ] = \frac{\text{density}}{Mw}$

density and molar mass of pure solids and pure liquids remains constant regardless of amount present.

76) $2H_2S \rightleftharpoons 2H_2 + S_2$

5 mole 1 mole 4mole

$K_C = \frac{1^2 \times 4}{(5)^2} = \frac{4}{25}$

77)

$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

$$K_p = \frac{(P_{NH_3})^1 (P_{H_2S})^1}{(1)}$$

78) **Explain Question:** For the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) +$ heat , find the relation between Kp and Kc .

Concept: $Kp = Kc(RT)^{\Delta n}$

Solution:

Δn = (moles of gaseous products) - (moles of gaseous reactants).

$\Delta n = 2 - 4 = -2$

$Kp = Kc(RT)^{-2}$

Final Answer: Option 3.

79)

Unit of $K_p \rightarrow (atm)^{\Delta n_g}$

$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$

$\Delta n_g = (1+3) - (2) = 2$

Unit = $(atm)^2$

80)

For $\Delta n_g < 0$

$K_p < K_c$

For $2SO_2 + O_2 \rightleftharpoons 2SO_3$

$\Delta n_g = (2) - (2 + 1) = -1$

$\Rightarrow K_p < K_c$

81)

Explanation:

(a) is incorrect; the modern periodic table is based on atomic number.

(b) is correct; the first d-block series ranges from Sc to Zn.

(c) is correct; alkali and alkaline earth metals are s-block elements.

(d) is correct; chalcogens and halogens are p-block elements.

Therefore, the correct statements are (b), (c), and (d).

Final Answer : Option (3)

82) At no. difference is same between the sucessive members of each group.

83) **Explain Question:** Calculate atomic weight of intermediate element in Dobereiner's triad.

Concept: This question is based on Dobereiner's triad → middle atomic weight ≈ average of extremes

Solution: Atomic weight of Y = (Atomic weight of X + Atomic weight of Z)/2

= (40 + 60)/2 = 100/2 = 50

Final Answer: 2

84) **Explain Question:** In a 100-element table, how many have 4d electrons?
Concept: This question is based on electron configuration
Solution: The 4d subshell begins filling at atomic number 39 (Yttrium). In a table with 100 elements, all elements from 39 to 100 will have 4d electrons.
Final Answer: 2

85)

Boron exhibits a diagonal relationship with Silicon.

86)

Tantalum comes in 5th group.
V → 4th period
Nb → 5th period
Ta → 6th period
so period no. will be 6 & group no. will be 5th.

87)

Actinide series (90-103)
Atomic number just after actinide series is 104. And according to IUPAC 1 = un, 0 = nil, 4 = quad
Suffix = ium

88)

Gadolinium have exceptional configuration that is why its one electron goes in 5d.
So the configuration becomes [Xe] $4f^7$ $5d^1$ $6s^2$ according to increasing order of n.

89)

As from 4th period d subshell starts getting filled. And the lowest atomic no. element will be in starting. n will be 4 so $3d^6$ $4s^2$ will be the lowest atomic no. configuration.

90)

For s & p n will be same
for d → (n–1)
if period no. = 4 then n = 4
and if n = 4 then d = n–1 i.e. it will be 3d in fourth period & element of 4th period will be containing e^- in 3d not in 4d.

BIOLOGY

91) NCERT XI, Page # 24, Para-2.3.3

92) NCERT, Pg.# 21

93) NCERT XI, Pg # 19

94) NCERT-XI, Pg. # 19

95) NCERT-XI, Pg-23,24

96) NCERT XI Page # 26

97) NCERT (XIth) Pg. # 22

98) NCERT (XI) Pg. # 21

99) NCERT (XI) Pg. # 26

100)

Generated by Allie

Problem Statement: The question is asking about the nature of viroids, specifically what type of genetic material they have and whether it is enclosed by a protein coat. It provides several options related to DNA and RNA forms and asks which correctly describes viroids.

Underlying Concept: Viroids are small infectious agents that differ from viruses because they consist solely of a short strand of single-stranded RNA (ssRNA) and lack a protein coat that typically surrounds viruses. Understanding the structure of genetic material and its association with protein coats is key to differentiating viroids from viruses and other pathogens.

Tips and Tricks: Remember that viruses usually have their genetic material enclosed in a protein coat called a capsid, but viroids are unique because they have RNA alone without such a coat.

Common Mistakes: Confusing viroids with viruses and assuming viroids have a protein coat encapsulating DNA or RNA; assuming double-stranded RNA or DNA when viroids actually have single-stranded RNA.

Why Other Options Are Incorrect?: Options indicating DNA (either ss or ds) enclosed by a protein coat are incorrect because viroids do not contain DNA and lack a protein coat. The choice stating ds RNA enclosed by a protein coat is also incorrect, as viroids have single-stranded RNA and no capsid.

101)

Solution:

The correct answer is:

4. Cell membrane

The Five Kingdom system (Whittaker) is based on mode of nutrition, cell structure, and body organization, but cell membrane is not a primary factor in this classification.

102)

In Carl Woese's classification system, the kingdom that includes fungi is Eukarya.

The correct answer is: OPTION (2) Eukarya

103)

Explain Question: Question asks how many of the listed groups are photosynthetic.

Concept: This question is based on Protista

Solution: 3 groups are photosynthetic (Dinoflagellates, Diatoms, Euglenoids).
Final answer: 2

104) NCERT XI Pg # 22

105)

NCERT XI- Page # 20, 21

106) NCERT Pg. # 23, para 2.3.2

107) NCERT-XI, Pg. # 201

108)

Question Explanation : Question is asking to correctly match different groups of Protists (Column I) with their representative examples (Column II).
Concept: Protista
Solution: (a) Chrysophytes → (iv) Diatoms
(b) Dinoflagellates → (iii) Gonyaulax
(c) Euglenoids → (ii) Euglena
(d) Protozoans → (i) Paramoecium
Final answer: Option 4

109)

Question Explanation: Asking to Identify the given figure and select the correct statements regarding it ?

Concept: Euglena

Solution: The figure is Euglena (elongated body with a flagellum).



(ii) Euglena has one functional flagellum → Correct

(iii) Euglena contains chlorophyll pigments similar to higher plants → Correct

(i) Incorrect → it is not Paramecium

(iv) Incorrect → Paramecium is ciliated, but this organism is flagellated

Final answer: 4(ii) and (iii).

110)

Question Explaination : Question is asking to identify the incorrect statement about fungi.
Concept: Fungi- General characters and fungal classes
Solution: Mushroom → Basidiomycetes, not Ascomycetes.-wrong
(true).plasmogamy → karyogamy → meiosisIn fungi,
Yeast → unicellular, non-mycelial fungus (true).
Fungi bodies → made of hyphae (true).
Final answer: Option 1

111) **Explain Question:** Question asks about which statements about sexual reproduction are true for Figure A (Aspergillus) and Figure B (Agaricus).

Concept: Ascomycetes and Basidiomycetes

Solution: Figure A -Aspergillus Figure B -Agaricus

Statement 1: Sexual reproduction in A occurs by ascospores → True (Aspergillus = Ascomycetes).

Final answer: 1

112) **Explain Question**: Question asks about judging the truth of an assertion–reason pair and whether the reason explains the assertion.
Concept: Fungi general characters and sexual reproduction in fungi

Solution: (A)Fungi show great diversity in morphology (yeasts, moulds, mushrooms) and habitat (soil, water, decaying matter, symbiotic forms).-correct

(R)Sexual cycle in fungi universally includes plasmogamy → karyogamy → meiosis, but this sequence does not explain their morphological and habitat diversity.

Final answer: 2

113)

Ncert page 17.

114)

NCERT XI Pg. # 26

115) **Explanation :-** The question asks to identify the correct characteristic of prions.
Concept :- This question is based on Prions
Solution :- Abnormal folded protein - Correct.
Prions are misfolded proteins that can induce other proteins to misfold, leading to disease.
Answer :- 4

116)

Explain Question : The question asks for the correct names and nutritional types of the algal and fungal partners in a lichen.

Concept : Biological Classification → Kingdom Fungi → Lichens

Solution :

A. Algal component → Phycobiont; autotrophic

B. Fungal component → Mycobiont; heterotrophic

Final answer : 4

117) NCERT XI Pg 23, 24

118)

Question Explanation: Question asks about the truth of an assertion–reason pair regarding lichens and their role in pollution indication.

Concept: Lichens

Solution: (A)Lichens are **sensitive to pollution**, especially **SO_2**, so they **cannot grow in polluted areas**.

(R)Their absence in polluted areas **explains why they are good pollution indicators.**

Reason correctly explains the assertion.

Final answer: Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

119)

Solution/Explanation :
Bacteriophages (or phages) are viruses that infect bacteria. Their structure typically consists of **proteins** (which form the capsid) and **genetic material** (either DNA or RNA). The genetic material of most bacteriophages is **double-stranded DNA (ds DNA)**.

120)

NCERT-XI, Pg. # 10

121)

Generated by Allie

Problem Statement: The question asks us to identify which one of the given options correctly groups examples under the three biological kingdoms: Monera, Protista, and Plantae. We need to match specific organisms to their correct kingdom categories.

Underlying Concept: The classification of living organisms into different kingdoms is based on their cellular structure, mode of nutrition, reproduction, and other biological characteristics. Monera includes prokaryotic organisms like bacteria; Protista includes eukaryotic unicellular or simple multicellular organisms; Plantae includes multicellular, primarily photosynthetic plants.

Tips and Tricks: Remember that blue-green algae (cyanobacteria) and mycoplasma are bacteria, so they belong to Monera. Slime molds are unique but classified under Protista and not fungi or plants. Solanum and Mangifera are common plants, thus in Plantae.

Common Mistakes: One common mistake is confusing slime moulds as fungi or plants rather than placing them under Protista. Another is grouping fungi under Plantae, which is incorrect.

Why Other Options Are Incorrect?: Options containing fungi such as Mucor and yeast under Plantae or Monera are incorrect since fungi belong to a separate kingdom. Grouping slime moulds with Monera or Plantae is also wrong since they belong to Protista.

122) NCERT XI Pg. 17

123) **Explanation :**
Statement based question - Choose correct option

Concept :

Kingdom Protista

Solution :
Statement I - True
Protista are unicellular or simple eukaryotic organism
Statement II - False
Because protist are characterical by well defined nucleus and other membrane - bound organelles.

Final Answer :
Option (3)

124)

Let's match the epithelial tissues with their locations:

A. (A) Cuboidal: Cuboidal epithelium is often found in ducts and tubules where secretion and absorption take place. The lining of the pancreatic duct (5) is a good example.

B. (B) Ciliated: Ciliated epithelium has cilia that help move substances across the cell surface. The inner lining of the Fallopian tube (4) is a classic example, where cilia help move the ovum.

C. (C) Brush-Border Columnar: Brush-border columnar epithelium has microvilli that increase surface area for absorption. The inner surface of the gall bladder (3) is a good example.

D. (D) Squamous: Squamous epithelium is thin and flat, ideal for diffusion. The inner lining of blood vessels (2) is made of simple squamous epithelium (endothelium).

E. (E) Keratinized squamous: Keratinized squamous epithelium is tough and protective. The epidermis of the skin (1) is made of keratinized stratified squamous epithelium.

The correct option is 3. (A)-(5), (B)-(4), (C)-(3), (D)-(2), (E)-(1)

125) **Explain Question :** Fill up the blanks by choosing correct combination of A, B, C and D
Concept : This question is based on Animal tissues – epithelial tissues and glands
Solution :
1. Single thin layer of flattened cells → Squamous epithelium (A)
2. Digestive enzymes secreted by → Exocrine glands (B)
3. Protection against chemical & mechanical stress → Compound epithelium (C)
4. Cell junctions are found in → Epithelial tissue (D)
Final answer : Option 2.

126) NCERT (Eng.) Pg. # 101, Para no. 6

127) Module-5, Page#95,99

128) **Question Explanation:** Match the correct tissue names(A,B,C) based on their structure and function as shown in figure.
Concept: Connective tissue
Solution:
(A) Beneath skin, helps its attachment to underlying muscle--Areolar connective tissue
(B) Connects muscle to bone--Tendon
(C) Connects bone to bone--Ligament

Final Answer: Option(4)

129)

A. A. Most abundant formed elements of blood: The most abundant formed elements in blood are erythrocytes (red blood cells). So, A matches with vi.

B. B. Collagen of areolar tissue is produced by: Collagen fibers in areolar tissue are produced by fibroblasts. So, B matches with vii.

C. C. Tendons are composed of this tissue: Tendons are composed of dense regular connective tissue. So, C matches with iii.

D. D. Provide shape to external ear: The external ear gets its shape from elastic cartilage. So, D matches with iv.

Therefore, the correct matching is:

A. A - vi, B - vii, C - iii, D - iv

The correct option is 4. A-vi B-vii C-iii D-iv.

130) NCERT Page # 103, 104

131)

Generated by Allie

Problem Statement: The question asks which option is incorrect when describing the characteristics of areolar tissue and adipose tissue. You need to identify the option that does not apply to both these types of tissues.

Underlying Concept: Areolar tissue and adipose tissue are types of connective tissue in the body. Areolar tissue is a loose connective tissue that holds organs together and provides

cushioning, while adipose tissue stores fat and is found beneath the skin. Both tissues belong to the category of loose connective tissues, not dense connective tissues.

Tips and Tricks: Remember that 'loose' connective tissues have more ground substance and fewer fibers, providing flexibility and cushioning; both areolar and adipose tissues fall under this category.

Common Mistakes: Confusing loose connective tissues with dense connective tissues, or mixing up the locations where these tissues are found.

Why Other Options Are Incorrect?: 'Loose connective tissue' and 'Present beneath the skin' are correct characteristics of both areolar and adipose tissues. Therefore, these options are not incorrect. The option 'Dense connective tissue' is incorrect because neither areolar nor adipose tissue is classified as dense connective tissue.

132) **[NCERT Pg :101-102]**

133) NCERT PAGE : 102

134) NCERT-XII, Pg. # 104

135) NCERT-XII, Pg. # 102

136) Module-4

137) Module-4

138) NCERT XI (E), Page No. 104

139) NCERT-XII, Pg. # 102

140)

Correct Answer:

3. b, d and e
Explanation:

Let's evaluate each statement:

A. (b) Cell junctions provide only structural links between its individual cells.

A. This is incorrect. While cell junctions provide structural links, they also play

important roles in communication (e.g., gap junctions) and regulating the movement of substances (e.g., tight junctions).

B. (d) Cuboidal epithelium is found in the thin part of the loop of Henle.

A. This is incorrect. The thin part of the loop of Henle is lined by simple squamous epithelium, not cuboidal epithelium.

C. (e) All glands are composed of both epithelial and connective tissue.

A. This is incorrect. While glands are derived from epithelial tissue, they typically also have surrounding connective tissue, but not all glands have both in the same proportion. For example, endocrine glands primarily consist of epithelial tissue.

141)

NCERT - XI. Pg # 101

142) **Explain Question:**
Ciliated epithelial cells are known to occur in -

Concept:
This question is based on Epithelial tissue - ciliated epithelium and its location.

Explanation :
Ciliated epithelium occurs in trachea and bronchioles, where cilia help move mucus and trapped particles.

Final answer : 2

143) NCERT XIIth Pg # 286

144) **Solution :** Alcium ions play a very important role in clotting. The enzyme complex thrombokinase involves a series of reactions requiring calcium ions for the conversion of prothrombin to thrombin.
Final Answer : (3)

145) **Solution :** In subsequent pregnancies, Rh antibodies from the Rh-negative mother can leak into the blood of the Rh-positive foetus and destroy the foetal RBCs, leading to erythroblastosis foetalis.
Final Answer : (2)

146) **Solution :** Blood group 'O' has no antigens on RBCs (nil), so it does not induce clumping

when donated to other groups, making 'O' individuals universal donors.
Final Answer : (2)

147) **Solution :** QRS complex represents depolarisation of ventricles, initiating ventricular contraction; P-wave responsible for atrial depolarisation.
Final Answer : (3)

148) **Solution :** a–iii, b–iv, c–ii, d–i, e–v
Neutrophils 60-65%, Eosinophils 2-3%, Basophils 0.5-1%, Lymphocytes 20-25%, Monocytes 6-8%.
Final Answer : (3)

149) **Solution :** "A unique vascular connection exists between the digestive tract and liver called hepatic portal system. The hepatic portal vein carries blood from intestine to the liver..." (implies processing nutrients/toxins in liver).
Final Answer : (3)

150) **Solution :** Inner tunica intima, middle tunica media, external tunica externa.
Final Answer : (2)

151) **Solution :** Cardiac output as stroke volume (70 mL) times heart rate (72/min), averaging 5000 mL or 5 litres.
Final Answer : (1)

152) **Solution :** Angina chest pain from low oxygen to heart; common in middle-aged, elderly, but can occur any age.
Final Answer : (2)

153) **Solution :** a–iv, b–iii, c–ii, d–i
A = anti-B, B = anti-A, AB = nil, O = anti-A, B.
Final Answer : (4)

154) **Solution :** Amphibians have 3-chambered heart with two atria and single ventricle, leading to mixed blood (incomplete double circulation).
Final Answer : (2)

155) **Solution :** RBCs have a lifespan of “about 120 days”, not 60 days.
Final Answer : (3)

156) **Solution :** The pulmonary valve prevents backflow into the right ventricle, not the left ventricle.
Final Answer : (3)

157) **Solution :** "Cardiac output = stroke volume × heart rate", and it varies based on activity level.
Final Answer : (2)

158) **Solution :** Globulins are plasma proteins that function in immune responses, including antigen-antibody reactions.
Final Answer : (3)

159) **Solution :** Basophils release histamine, which plays a role in allergic responses and inflammation.
Final Answer : (2)

160) **Solution :** ABO grouping is the classification of human blood based on the inherited properties of red blood cells (erythrocytes). It is determined by the presence or absence of the antigens A and B, which are carried on the surface of the red cells. Persons may thus have type A, type B, type O, or type AB blood. The A, B, and O blood groups were first identified by Austrian immunologist Karl Landsteiner in 1901.
Final Answer : (4)

161) **Solution :** LUBB is the first heart sound, which is associated with the closure of the tricuspid and bicuspid valves at the beginning of systole.
Final Answer : (3)

162) **Solution :** ECG (Electrocardiogram) is a graphic sketching of the variations in electrical potential caused by the excitation of the heart muscle and detected at the body surface. The normal ECG is a scalar representation that shows deflections resulting from cardiac activity. It comprises the P wave, QRS complex, T wave and U wave. QRS complex: It represents the depolarization of the ventricles which initiates the ventricular contraction. The contraction starts shortly after Q and marks the beginning of the systole. Hence QRS complex shows one complete pulse.
Final Answer : (3)

163) **Solution :** The atrioventricular septum is a septum of the heart between the right atrium (RA) and the left ventricle. It is a small part of the membranous septum of the heart just above the septal cusp of the tricuspid valve.
Final Answer : (2)

164) NCERT Pg. # 202

165) NCERT Pg. # 196

166) NCERT Pg. # 286, 287 (18.4 & 18.5)

167) NCERT Pg. # 284, (18.3.2)

168) NCERT Pg. # 284, (18.3.1)

169) **Question Explanation :**
Which protein is involved in coagulation of blood?

Concept :
This question is based on Blood clotting.

Solution :
Fibrinogen is involved in blood clotting.

Final Answer : (3)

170) NCERT Pg#279-280 (E), 280-281 (H)

171)

Question Explanation: The question is asking which blood types a person with type A Rh- blood can safely receive in a blood transfusion

Concept: Blood group/Rh factor.

Solution:
A person with A Rh- blood has A antigens and no Rh antigen, so they cannot receive B, AB, or Rh+ blood.

They can receive only from donors who are A Rh- or O Rh-.

Final answer: Option (3) iv and v

172)

NCERT Pg # 288

173)

NCERT Based.
SA node (wave of contraction get originated at SA node)
↓
AV node
↓
Bundle of His
↓
Purkinje's fibre (now, the impulse stimulates the ventricles to contract simultaneously)

174) Because the origin of all muscle except ciliary muscle, has mesodermal origin.

175)

NCERT XI Pg # 278

176) NCERT pg no.283

177) NCERT pg no.286 + 287

178) **Question explanation**: To determine the correct sequence of heart valves through which blood flows from vena cava.
Concept: Blood circulation through heart.
Solution: Blood enters right atrium from vena cava → passes through tricuspid valve → right ventricle → pulmonary semilunar valve → lungs → left atrium → bicuspid (mitral) valve → left ventricle → aortic semilunar valve.
Final Answer: Option (1) is correct.

179) NCERT pg no.285

180)

Question Explanation
The question is asking what happens in the heart during ventricular systole.

Concept
Cardiac cycle – ventricular systole.

Solution
During ventricular systole, the ventricles contract, closing the tricuspid and bicuspid (atrioventricular) valves to prevent backflow, while the semilunar valves open to pump blood into the arteries. Ventricular pressure rises, not declines, and nearly 70 percent of the blood is pumped out, not 30 percent.

Final answer
Option 3